

ABCDs of Nutritional Evaluation

Part C: Clinical Findings-Nutrition Physical Exam



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APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Elizabeth Boham, MD, RD, disclosed no relevant financial relationships with any ineligible companies.

Michael Stone, MD, is a consultant for Tailored Script and Calroy Health.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Explain the rationale for performing a functional nutrition physical exam.
2. List the parts of the physical exam that provide the highest yield for detecting nutrient insufficiencies.
3. Describe the difference between nutrient deficiencies and insufficiencies and its relevance vis-à-vis the functional nutrition physical exam.
4. Recall common patterns of nutrition exam findings for various nutrients.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Perform aspects of a functional nutrition physical exam to detect nutrient deficiencies and insufficiencies.

Clinical Focus

- Overview and rationale for the functional nutrition physical exam.
- Highest yield body tissues for detecting nutrient insufficiencies.



ABCDs of the Functional Nutrition Evaluation



Anthropometrics &
Vital Signs



Biomarkers &
Functional Labs

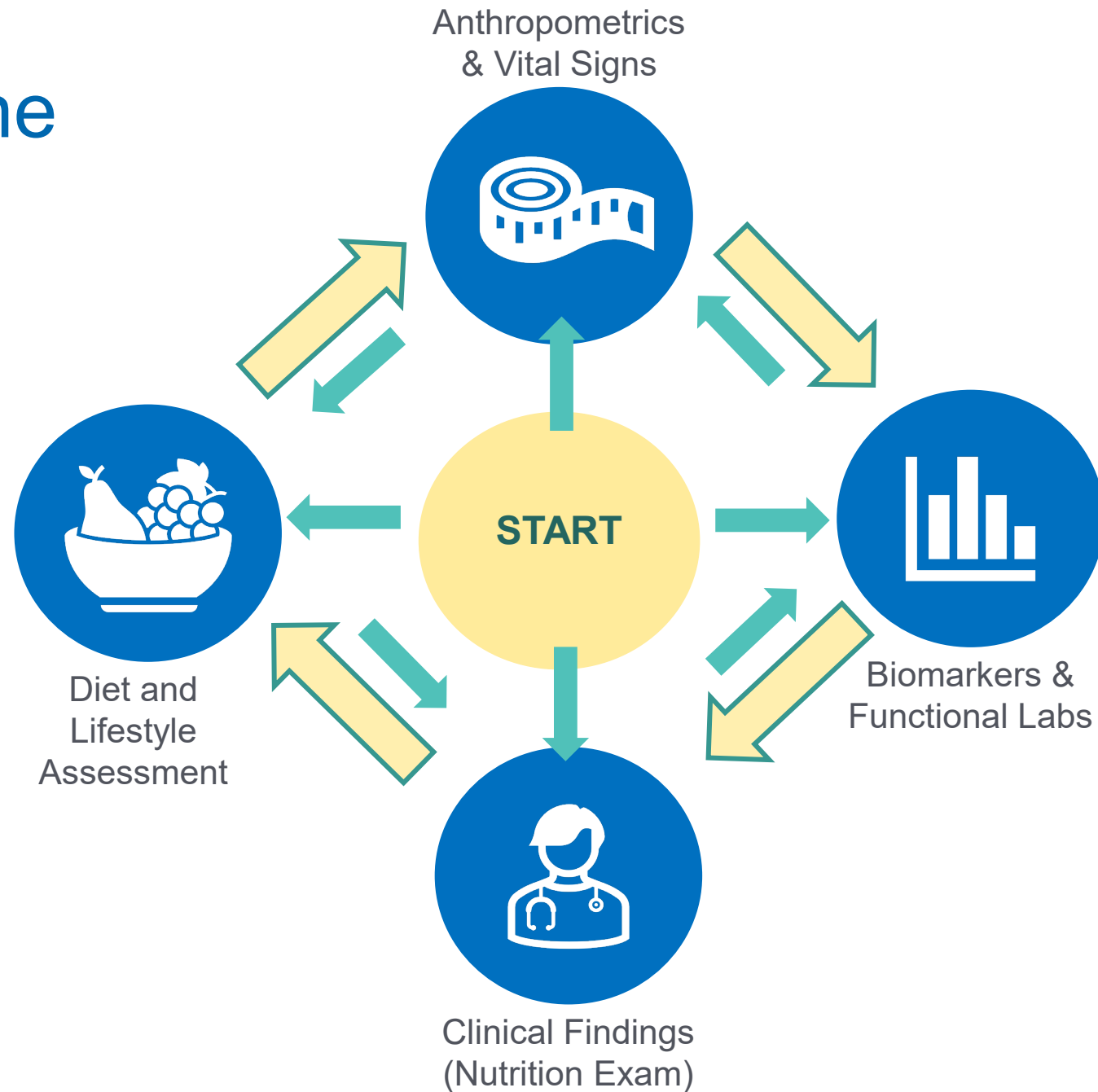


**Clinical Findings
(Nutrition Exam)**



Diet and Lifestyle
Assessment

ABCDs of the Functional Nutrition Evaluation



Rationale for the Functional Nutrition Physical Exam

Why perform a “functional nutrition physical exam” for insufficiencies?

How does it differ from a “nutrient deficiency focused” exam?



Images courtesy of Michael Stone, MD

Rationale for the Functional Nutrition Physical Exam

- High prevalence of nutrient deficiencies and insufficiencies.
- Functional nutrition exam rarely taught in conventional training.
- Effectiveness of nutrient repletion in chronic diseases.
- Judicious use of laboratory testing.
 - “Choosing Wisely”



Images courtesy of Michael Stone, MD

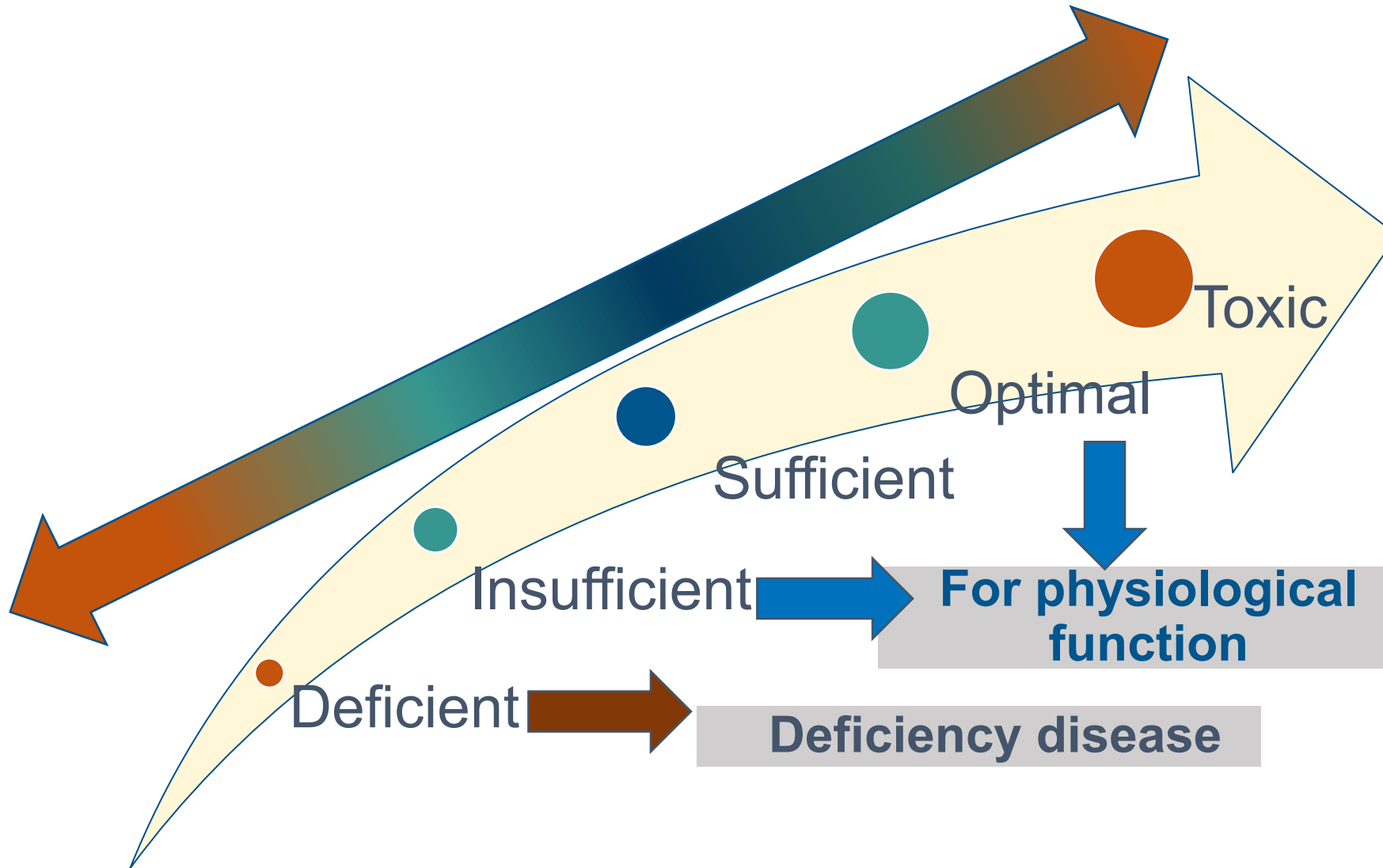
Rationale for the Functional Nutrition Physical Exam

- Identification of insufficiencies that mediate dysfunction, not just deficiency diseases.
- Develops patient trust and therapeutic partnership.
- Patients are highly motivated by personal physical findings.
- Ability to track progress and improvement.



Images courtesy of Michael Stone, MD

CONTINUUM OF NUTRIENT NEEDS



Long Description: Continuum of Nutrient Needs.

Breakdown as follows:

- **Deficient:** This starts on the left, indicating a lack of necessary nutrients, which may lead to deficiency diseases.
- **Insufficient:** Indicates a level where nutrients are present but not enough for optimal health.
- **Sufficient:** A level where the nutrient intake is adequate for basic physiological function.
- **Optimal:** The range where nutrient intake supports the best physiological function and health.
- **Toxic:** On the far right, this indicates excessive nutrient intake, which can lead to toxicity.
- The diagram uses an arrow to show the range and highlights that moving too far in either direction (deficient or toxic) can have negative health impacts.

In Adults Ages 19-71 Years Old

Inadequate Intake From Food or Food + Supplements

- <10% Copper, Iron, Selenium, Phosphorus, Thiamin, Riboflavin, Niacin, B12
- 10-50% Zinc, Vitamin A, Folate, B6, Vitamin C
- 50-75% Calcium, Magnesium
- 75-95% Vitamin D, Vitamin E



Vitamin	Food Source	Role/Function
Thiamine	Fresh vegetables, husk of cereal grains, meats, liver	Energy metabolism, synthesis of myelin
Riboflavin	Milk and meat products	Energy metabolism
Nicotinic acid	Liver, lean meats, cereals, legumes	Energy metabolism
Pyridoxine	Meats, cereals, lentils, nuts, some fruit & vegetables	Amino acid metabolism
Pantothenic acid	Liver, meats, cereals, milk, egg yolk, vegetables	Energy metabolism
Folate	Yeast, liver, fresh green vegetables	Nucleotide metabolism
B12	Meat, fish, poultry, milk/products, colon bacteria	Nucleotide metabolism
C	Fresh fruits and vegetables	Synthesis of connective tissue & repair, antioxidant
A	Animal tissue, liver, green plants	Vision pigment, immune balance
D	UV light, mushrooms, added to dairy	Controls calcium uptake into the body, bone
E	Vegetable seed oils, milk, eggs, meat	Antioxidant
K	Green leafy vegetables, meats, dairy	Activation of blood clotting factors
Biotin	Yeast, organ meats, muscle meats, dairy products	Amino acid metabolism
Choline	Meat, egg yolk, legumes, cereals	Neurotransmitter, phospholipid component



Female adults with underconsumption



Both male and female adults have significant underconsumption

Mineral	Food Source	Role/Functions
Calcium	Milk, dairy, grains, green vegetables, fruit	Bone, teeth rigidity, blood clotting metabolism
Iron	Most animal and plant foods	>300 metabolic reactions, energy, CYP450, oxygen transport, DNA/RNA structure, thyroperoxidase
Magnesium	Vegetables, fruit, grains, meat, dairy products	>300 metabolic reactions, neuron, muscle, ATP/ADP metabolism
Zinc	Red meats, shellfish, wholegrain cereals, legumes, nuts	3,000 reactions, metabolic reactions, RNA/DNA stabilization, ribosomes, immune function
Copper	Liver and organ meats, grains, legumes, nuts, seeds, cocoa	Component of many enzymes, cytochrome C oxidases, amine oxidases, lysine oxidase
Selenium	Organ meats, seafood, muscle meats, cereals and grains	Glutathione peroxidase, lodothyronine deiodinase, thioredoxin reductase
Manganese	Unrefined cereals, nuts, leafy vegetables, tea	Glycosyl transferase, arginase, pyruvate carboxylase
Molybdenum	Milk, milk products, dried legumes/pulses, liver	Aldehyde oxidase, xanthine oxidase, electron T. chain
Phosphorus	Meat, grains	Phosphate in all energy reactions in the cells
Iodine	Shellfish, cold water fish, liver, pineapple, eggs, peanuts, cheese, lettuce, spinach, green peppers	Production of T3, T4



Female adults with underconsumption



Both male and female adults have significant underconsumption

Identifying Nutrient Insufficiencies

Highest Yield Body Tissues

Tissues with rapid turnover or metabolic vulnerability:

- Hair
- Skin
- Nails
- Mucosa (mouth, including taste)



Images courtesy of Michael Stone, MD

Cells Vary in Their Turnover

- **Tongue (taste buds):** Days to 2 weeks (14 days)
- **Mucosa (mouth, nasopharynx, bowel):** Days to 2-3 weeks
- **Skin:** 4-7 weeks (28-49 days)
- **Nails:** 4-24 weeks (28-168 days), toenails up to 450 days
- **Hair (growth phase):** Up to 1,000-2,000 days

Clinical Focus

- Hair Exam and Nutrient Associations



The ABCDs of Nutrition Evaluation Hair Exam



Hair Exam

DISTRIBUTION – COLOR – TEXTURE

Alopecia



Discolored



Images courtesy of Michael Stone, MD

Start With the Hair and the Scalp

- Change in Color
- Change in Texture
- Change in Thickness
- Change in Distribution
- Thinning or Loss

Hair on the Scalp:

- Quantity and Distribution:
 - Thinning, alopecia, male pattern balding.
- Quality:
 - Coarseness, thinning, luster, color change.
- Scalp:
 - Skin changes, lesions, hidden growths – from the crown and all around.



Image courtesy of Michael Stone, MD

Look at the temples
Hairline receding
Part widening



Images courtesy of Michael Stone, MD

Widening, less distinct part, note loss of hair density



Androgenic pattern hair loss –
look for functional nutrition inadequacy

Images courtesy of Michael Stone, MD

Female Pattern Alopecia – Hair Loss



Bitemporal Thinning



Ludwig 1 Mild Thinning



Ludwig 2 Widened Part



Ludwig 3 No Part



Hair & Scalp Findings

Nutrient Associations

- **Dry hair:** essential fatty acids (EFAs)
- **Dandruff:** EFAs, zinc
- **Dull hair:** protein, iron
- **Premature gray:** copper, B12
- **Depigmentation:** protein, copper, zinc, biotin, B12
- **Corkscrew hair:** vitamin C
- **Swan neck hair:** vitamin C
- **Menke's hair:** copper
- **Thinning hair:** protein, zinc, selenium



Condition-Specific Nutrition Insufficiencies

Alopecia

- **Protein** Inadequacy
 - Methionine, L cysteine, total protein
- **Minerals**
 - Zinc, copper, iron
- **Vitamins**
 - B complex, biotin
 - Vitamins C, E, A, D
- **CoQ10**



Macronutrient Associations With Hair and Scalp Findings

Protein and Fat

Protein	Physical Exam Findings
Protein deficiency	Alopecia, brittle/fragile hair, depigmentation of hair and skin, flag sign (bands of depigmentation), hair plucked easily, hypertrichosis (excessive hair growth), reddish-brown coloring, telogen effluvium, thinning

Fats & Oils	Physical Exam Findings
EFA deficiency	Alopecia, hair loss of scalp and eyebrows, lightening of hair (depigmentation), telogen effluvium

Micronutrient Associations With Hair and Scalp Findings

Minerals

Minerals	Physical Exam Findings
Copper deficiency	Brittle/fragile hair, depigmentation of hair and skin, Menke's disease (kinky hair disease)
Iron deficiency	Alopecia, telogen effluvium
Manganese deficiency	Slowed growth
Selenium deficiency	Alopecia, depigmentation of hair and skin (lightening of hair)
Selenium excess	Alopecia
Zinc deficiency	Alopecia, brittle/fragile hair, telogen effluvium, white hair

Micronutrient Associations With Hair and Scalp Findings

Vitamins

Vitamins	Physical Exam Findings
Vitamin A deficiency	Follicular hyperkeratosis
Vitamin A excess	Alopecia, telogen effluvium
Vitamin D deficiency	Alopecia
Niacin (B3) deficiency	Alopecia
Biotin (B7) deficiency	Alopecia, loss of hair on scalp, eyebrows, and eyelashes, trichorrhexis nodosa (nodular ball and socket deformities)
Folate (B9) deficiency	Depigmentation (lightening of hair)
Vitamin B12 deficiency	Depigmentation (lightening of hair), gray hair
Vitamin C deficiency	Corkscrew hairs, curling, follicular hyperkeratosis, follicular plugging, perifollicular hemorrhage, swan neck hairs

See: "References: Hair and Scalp Findings and Their Nutrient Associations"

References

HAIR AND SCALP FINDINGS AND THEIR NUTRIENT ASSOCIATIONS

- Aşkın Ö, Uzunçakmak TKÜ, Altunkalem N, Tüzün Y. Vitamin deficiencies/hypervitaminosis and the skin. *Clin Dermatol*. 2021;39(5):847-857. doi:10.1016/j.clindermatol.2021.05.010
- Castelo-Soccio LA. Hair manifestations of systemic disease. *Curr Probl Pediatr Adolesc Health Care*. 2012;42(8):198-203. doi:10.1016/j.cppeds.2012.02.002
- Dawber RP. Integumentary associations of pernicious anaemia. *Br J Dermatol*. 1970;82(3):221-223. doi:10.1111/j.1365-2133.1970.tb12428.x
- Dawber RP, Gummer CL. The colour of the hair. In: Dawber R, ed. *Diseases of the Hair and Scalp*. 3rd ed. Oxford: Blackwell Science; 1997:397-416.
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Finner AM. Nutrition and hair: deficiencies and supplements. *Dermatol Clin*. 2013;31(1):167-172. doi:10.1016/j.det.2012.08.015
- Hammond KA. Nutrition-focused physical assessment: laboratory tests and medications. In: Mahan LK, Escott-Stump S, Raymond JL. *Krause's Food and the Nutrition Care Process*. 13th ed. Saunders; 2012:1076-1078.
- Swartz MH. Hair coarseness (dry, course and sparse hypothyroidism) loss of hair: anemia, heavy metal poisoning, hypopituitarism, nutritional diseases such as pellagra. In: Swartz. MH. *Textbook Of Physical Diagnosis*. 6th ed. Saunders Elsevier; 2010:144.
- Weinsier RL, Butterworth CE. Hair pluckability - protein undernutrition and its many causes (malabsorption, underconsumption, increased requirement. In: Weinsier RL, Butterworth CE. *Handbook of Clinical Nutrition: Clinician's Manual for the Diagnosis and Management of Nutritional Problems*. Mosby; 1981:30.

References

HAIR AND SCALP FINDINGS AND THEIR NUTRIENT ASSOCIATIONS

- Hollfelder B, Blankenburg G, Wolfram LJ, Höcker H. Chemical and physical properties of pigmented and non-pigmented hair ('grey hair'). *Int J Cosmet Sci*. 1995;17(2):87-89. doi:10.1111/j.1467-2494.1995.tb00112.x
- Hughes EC, Syed HA, Saleh D. *Telogen Effluvium*. StatPearls Publishing; 2024. Updated May 1, 2024. Accessed June 8, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK430848/>
- Sinclair RD. Healthy hair: what is it? *J Invest Dermatol Symp Proc*. 2007;12(2):2-5. doi:10.1038/sj.jidsymp.5650046
- Nishimura EK, Granter SR, Fisher DE. Mechanisms of hair graying: incomplete melanocyte stem cell maintenance in the niche. *Science*. 2005;307(5710):720-724. doi:10.1126/science.1099593
- O'Connor K, Goldberg LJ. Nutrition and hair. *Clin Dermatol*. 2021;39(5):809-818. doi:10.1016/j.clindermatol.2021.05.008
- Tobin DJ, Paus R. Graying: gerontobiology of the hair follicle pigmentary unit [published correction appears in *Exp Gerontol*. 2001;36(3):591-592]. *Exp Gerontol*. 2001;36(1):29-54. doi:10.1016/s0531-5565(00)00210-2
- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003
- Wood JM, Decker H, Hartmann H, et al. Senile hair graying: H₂O₂-mediated oxidative stress affects human hair color by blunting methionine sulfoxide repair. *FASEB J*. 2009;23(7):2065-2075. doi:10.1096/fj.08-125435
- Yan AC, Jen MV. Skin signs of pediatric nutritional disorders. *Curr Probl Pediatr Adolesc Health Care*. 2012;42(8):212-217. doi:10.1016/j.cppeds.2012.02.003

Clinical Focus

- Mouth Exam and Nutrient Associations: Tongue



The ABCDs of Nutrition Evaluation Mouth Exam





Functional Nutrition Evaluation

Exam Form: Mouth and Dental

Patient Name _____ Date _____

Provider _____

1 Jaw Movement

- ☐ Symmetric (No Pain)
- ☐ Asymmetric ☐ L ☐ R
- ☐ Auscultated crepitus or click
- ☐ Pain ☐ L ☐ R
- ☐ Mouth Opening
 - ☐ More than 4 cm
 - ☐ Less than 4 cm

2 Lips

- ☐ Normal
- ☐ Dry, cracking
- ☐ Angular cracks or sores
- ☐ Ulcerations
- ☐ Fissures
- ☐ Perioral rash
- ☐ Loss of lip borders
- ☐ Lesions
 - ☐ Plaques ☐ Macules ☐ Cysts
- ☐ Edema, angioedema
- ☐ Piercing
- ☐ Metal ☐ Plastic

3 Soft Palate, Hard Palate, Tonsils, Pillars

- ☐ Normal
- ☐ Cleft or oropharyngeal defects
- ☐ Bony Lesions
 - ☐ Torus Palatinus
 - ☐ Torus Mandibularis
- ☐ Soft Palate
 - ☐ Asymmetric movement
 - ☐ Uvula swelling ☐ Bifid
 - ☐ Uvula absent
- ☐ Lesions:
 - ☐ Stomatitis ☐ Ulcerations
 - ☐ Petechiae or bruising
 - ☐ Other _____

4 Soft Palate, Hard Palate, Tonsils, Pillars (cont.)

- ☐ Tonsils
 - ☐ Regressed ☐ Hypertrophy
 - ☐ Removed
- ☐ Breath
 - ☐ Halitosis ☐ Acetone
 - ☐ Putrid ☐ Feculent
 - ☐ Sweet ☐ Other _____

4 Tongue

- ☐ Normal
- ☐ Abnormal movement
 - ☐ Fasciculations
 - ☐ Other _____
- ☐ Enlarged
- ☐ Scalloping
- ☐ Color
 - ☐ Red ☐ Pale
 - ☐ Black ☐ White
- ☐ Coating
 - ☐ Brown ☐ Black
 - ☐ Yellow ☐ White
- ☐ KOH+ ☐ Y ☐ N
- ☐ Pain ☐ Y ☐ N
- ☐ Taste bud distribution and prominence
 - ☐ Normal ☐ Abnormal
 - ☐ Geographic Tongue
- ☐ Fissuring
 - ☐ Central ☐ Transverse
- ☐ Ankyloglossia (tongue tied)
- ☐ Lesions
 - ☐ Ulcerations
 - ☐ Cysts ☐ Masses
 - ☐ Leukoplakia
 - ☐ Tongue varicosities
 - ☐ Wharton duct blocked ☐ Y ☐ N
- ☐ Saliva pH
 - ☐ Less than 6.8
 - ☐ More than 6.8

5 Gums

- ☐ Normal
- ☐ Bleeding
- ☐ Friability
- ☐ Bruising
- ☐ Lesions
 - ☐ Macules ☐ Color _____ ☐ Size _____
 - ☐ Tenderness
- ☐ Gum line darkening
- ☐ Gingivitis
- ☐ Periodontal Disease
- ☐ Hyperplasia

6 Buccal Mucosa

- ☐ Normal
- ☐ Abrasions
- ☐ Lesions
 - ☐ Macules ☐ Color _____ ☐ Size _____
 - ☐ Papules
 - ☐ Plaques
 - ☐ Ulceration
 - ☐ Linea Alba
- ☐ Tattoos ☐ Amalgam ☐ Other
- ☐ Stenson's duct papilla
 - ☐ Normal ☐ Y ☐ N
- ☐ Xerostomia

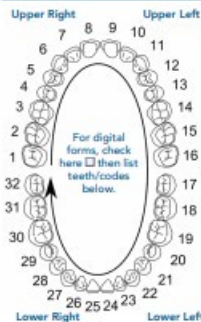
7 Teeth

- ☐ Healthy—No Restorations
- ☐ Missing Teeth
 - Numbers _____
- ☐ Tooth attrition or abrasions
- ☐ Silver/Mercury filling
 - Numbers _____
- ☐ Silver next to gold
 - Numbers _____
- ☐ Bridges/Dentures
- ☐ Periodontal ligament pain
- ☐ Aluminum foil pain or shocks
- ☐ Enamel marks
 - ☐ Dysplasia
- ☐ Discoloration
- ☐ Fluorosis
- ☐ Plaque (Tartar)

8 Chew/Swallow

- ☐ Normal swallowing
- ☐ Chew/swallow—Not checked
- ☐ Chew/swallow (cracker) without difficulty
- ☐ Difficulty swallowing (water)
- ☐ Chokes with swallowing
- ☐ O2 Sat drop more than 2%
- ☐ Taste:
 - Salty ☐ Y ☐ N
 - Sweet ☐ Y ☐ N
 - Bitter ☐ Y ☐ N
 - Sour ☐ Y ☐ N
- ☐ Taste Not Checked

U = Amalgam
GC = Gold Crown
PC = Porcelain Crown
PD = Periodontal Disease
RC = Root Canal
EX = Extraction



Referral to:
☐ Dentist ☐ Nutritionist
☐ ENT ☐ No Referral
☐ Oral Surgeon



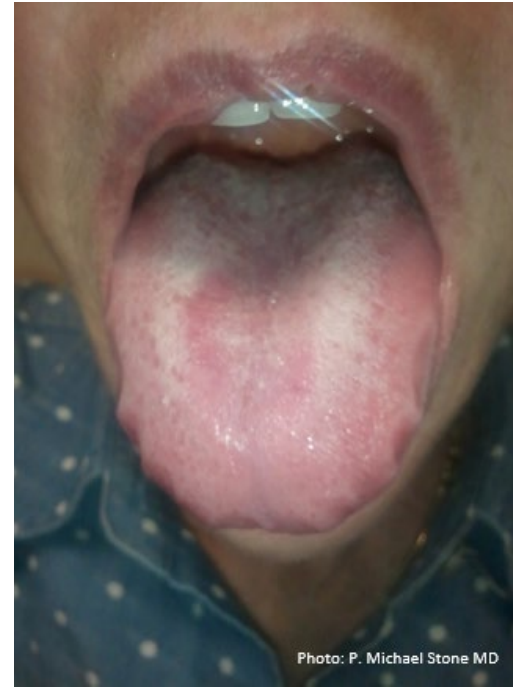
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Version 3

IFM Toolkit Asset Examination of the Mouth

Scalloping

- Many conditions have findings that are not known to be nutrient-related.
- Iodine deficiency, in association with hypothyroidism, has been documented.

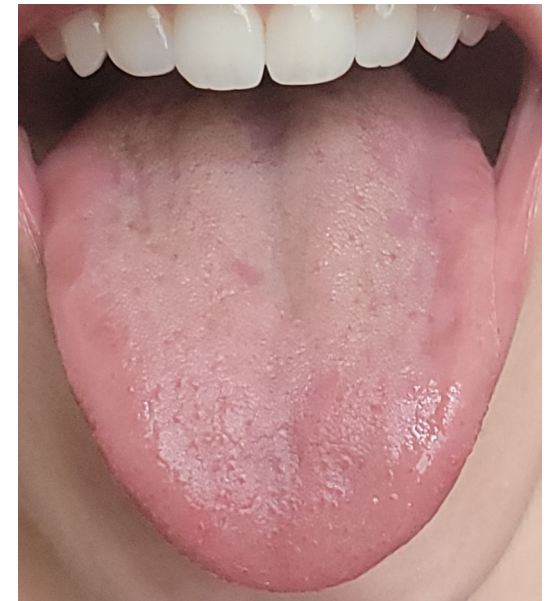


Images courtesy of Michael Stone, MD

- Costa SA, Brinhole MC, da Silva RA, Dos Santos DH, Tanabe MN. Surgical treatment of congenital true macroglossia. *Case Rep Dent*. 2013;2013:489194. doi:10.1155/2013/489194
- Mulla SA, Ansari A, Bhattacharjee M, Patil AH, Methal SA. Oral manifestations of nutritional deficiencies: a micro review of macro issues in the mouth. *Oral Maxillofac Pathol J*. 2023;14(2):217-219.
- Tolkachjov SN, Bruce AJ. Oral manifestations of nutritional disorders. *Clin Dermatol*. 2017;35(5):441-452. doi:10.1016/j.clindermatol.2017.06.009
- Reamy BV, Derby R, Bunt CW. Common tongue conditions in primary care. *Am Fam Physician*. 2010;81(5):627-634.

Geographic Tongue

- If **painful** and geographic:
 - Consider iron, folic acid, B2, B12, niacin, zinc insufficiencies



Images courtesy of Michael Stone, MD



From: Masferrer E, Jucgla A. Images in clinical medicine. Geographic tongue. *N Engl J Med*. 2009;361(20):e44. doi:10.1056/NEJMicm0810145 Copyright 2009 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

- Järvinen J, Kullaa-Mikkonen A, Kotilainen R. Some local and systemic factors related to tongue inflammation. *Proc Finn Dent Soc*. 1989;85(3):199-209.
- Reamy BV, Derby R, Bunt CW. Common tongue conditions in primary care. *Am Fam Physician*. 2010;81(5):627-634.
- Nandini DB, Bhavana SB, Deepak BS, Ashwini R. Paediatric geographic tongue: a case report, review and recent updates. *J Clin Diagn Res*. 2016;10(2):ZE05-ZE9. doi:10.7860/JCDR/2016/16452.7191

Geographic Tongue

- Frequency: 0.37-14.3% of pediatric population (up to 15%)
- 39% in people ages 20-29
- Females > Males 1.5:1
- **Nutrition:** Iron, zinc, niacin, B6, B12, folate insufficiency

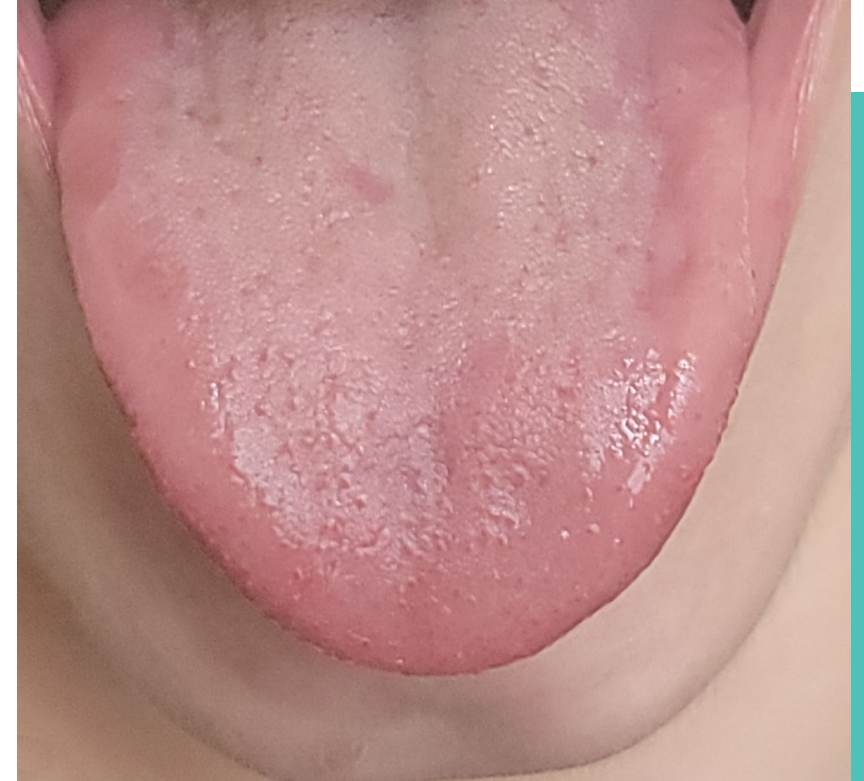


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- Nandini DB, Bhavana SB, Deepak BS, Ashwini R. Paediatric geographic tongue: a case report, review and recent updates. *J Clin Diagn Res.* 2016;10(2):ZE05-ZE9. doi:10.7860/JCDR/2016/16452.7191
- Esquivel-Pedraza L, Cicero-Casarrubias A, Fernández-Cuevas L, et al. Clinical atrophy patterns of filiform papillae of the tongue and their relationship with the serum levels of iron, vitamin B12 and folic acid in patients with systemic diseases. *Int J Dermatol.* 2023;62(7):915-923. doi:10.1111/ijd.16669
- Additional References: See "Geographic Tongue"

References

GEOGRAPHIC TONGUE

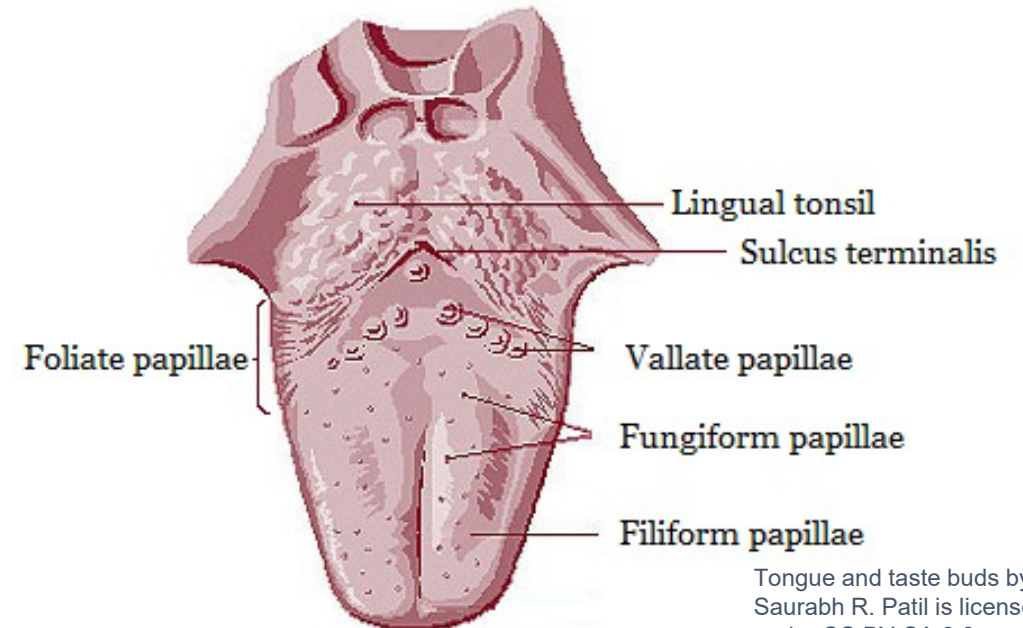
- Bačun B, Galić D, Pul L, Tomas M, Kuiš D. Manifestations and treatment of hypovitaminosis in oral diseases: a systematic review. *Dent J (Basel)*. 2024;12(6):152. doi:10.3390/dj12060152
- Khayamzadeh M, Najafi S, Sadrolodabaei P, Vakili F, Kharrazi Fard MJ. Determining salivary and serum levels of iron, zinc and vitamin B₁₂ in patients with geographic tongue. *J Dent Res Dent Clin Dent Prospects*. 2019;13(3):221-226. doi:10.15171/joddd.2019.034
- Järvinen J, Kullaa-Mikkonen A, Kotilainen R. Some local and systemic factors related to tongue inflammation. *Proc Finn Dent Soc*. 1989;85(3):199-209.
- Reamy BV, Derby R, Bunt CW. Common tongue conditions in primary care. *Am Fam Physician*. 2010;81(5):627-634.
- Nandini DB, Bhavana SB, Deepak BS, Ashwini R. Paediatric geographic tongue: a case report, review and recent updates. *J Clin Diagn Res*. 2016;10(2):ZE05-ZE9. doi:10.7860/JCDR/2016/16452.7191

Taste Bud and Papillae Distribution

- Circumvallate, meaning a battlement surrounded by a moat (inverted V)
- Foliate: leaf-shaped
- Filiform or thread-shaped
- Fungiform: mushroom-shaped (spots)



Image courtesy of Michael Stone, MD



Tongue

Glossitis With Taste Bud Atrophy

Nutritional Deficiency:

Protein Under-Nutrition, Iron,
Riboflavin, Niacin, B6, Folate,
Vitamin B12, Zinc



Rivadeneira, A., Moyer, P., Saliccioli, J. D.,
Pellagra in the USA: unusual manifestations
of a rare entity, BMJ Case Reports
CP2019;12:e230972.



Images Courtesy of Michael Stone, MD



Images Courtesy of Michael Stone, MD

References

GLOSSITIS

Infection:

- Chiang CP, Chang JY, Wang YP, Wu YH, Wu YC, Sun A. Atrophic glossitis: etiology, serum autoantibodies, anemia, hematinic deficiencies, hyperhomocysteinemia, and management. *J Formos Med Assoc.* 2020;119(4):774-780. doi:10.1016/j.jfma.2019.04.015

Irritation:

- Sharabi AF, Winters R. *Glossitis*. StatPearls Publishing; 2023. Updated August 8, 2023. Accessed June 8, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK560627/>

Allergic Reactions:

- de Groot A. Contact allergy to (ingredients of) toothpastes. *Dermatitis.* 2017;28(2):95-114. doi:10.1097/DER.0000000000000255

Iron Deficiency:

- Strączek A, Szałkowska J, Sutkowska P, et al. Impact of nutrition on the condition of the oral mucosa and periodontium: a narrative review. *Dent Med Probl.* 2023;60(4):697-707. doi:10.17219/dmp/156466

B12 Deficiency:

- Brescoll J, Daveluy S. A review of vitamin B12 in dermatology. *Am J Clin Dermatol.* 2015;16(1):27-33. doi:10.1007/s40257-014-0107-3

Folic Acid Deficiency:

- Chiang CP, Chang JY, Wang YP, Wu YH, Wu YC, Sun A. Atrophic glossitis: etiology, serum autoantibodies, anemia, hematinic deficiencies, hyperhomocysteinemia, and management. *J Formos Med Assoc.* 2020;119(4):774-780. doi:10.1016/j.jfma.2019.04.015
- Reamy BV, Derby R, Bunt CW. Common tongue conditions in primary care. *Am Fam Physician.* 2010;81(5):627-634.
- Neville BW, Damm DD, White DK. *Color Atlas of Clinical Oral Pathology*. 2nd ed. Williams & Wilkins; 2003.
- Rivadeneira A, Moyer P, Saliccioli JD. Pellagra in the USA: unusual manifestations of a rare entity. *BMJ Case Rep.* 2019;12(9):e230972. doi:10.1136/bcr-2019-230972

Fissuring – Longitudinal (Midline, Lateral)

Transverse, Lambda



Image courtesy of Michael Stone, MD



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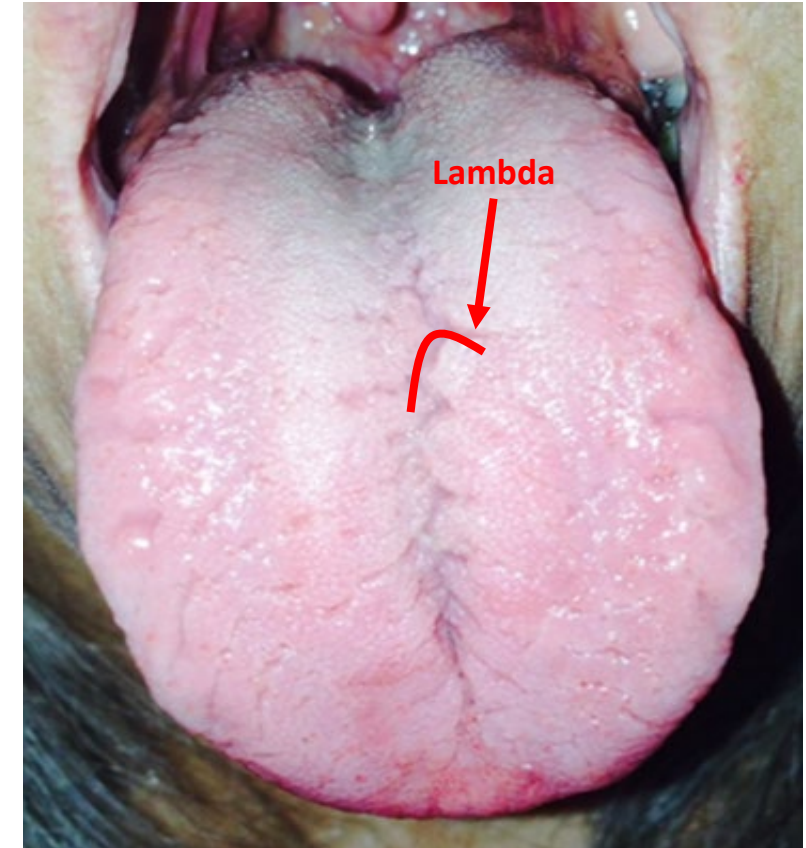


Image courtesy of Patrick Hanaway, MD

Fissuring

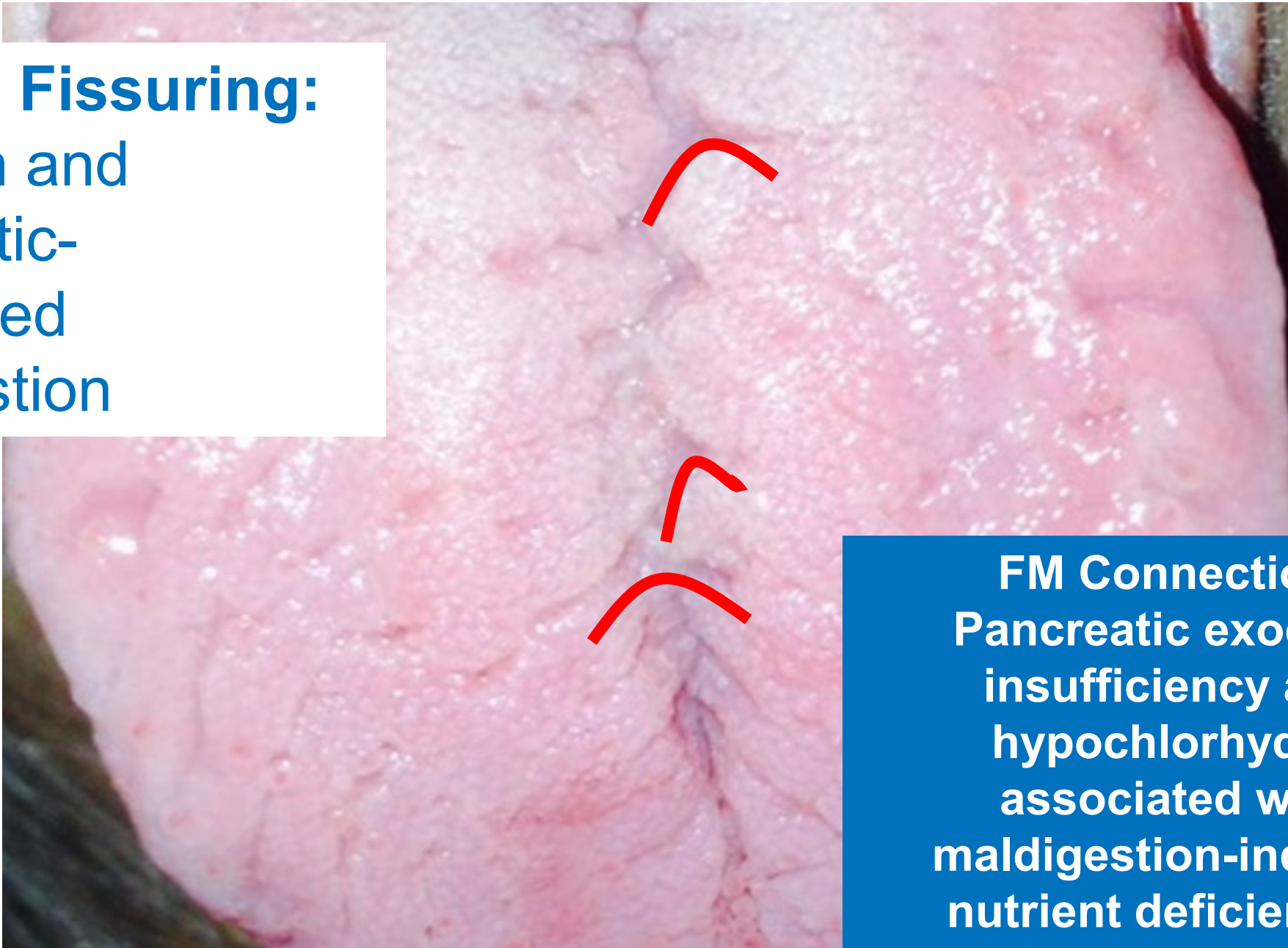
Nutritional Deficiency: B2, B12, Folate



Image courtesy of Michael Stone, MD

- Tolkachjov SN, Bruce AJ. Oral manifestations of nutritional disorders. *Clin Dermatol*. 2017;35(5):441-452. doi:10.1016/j.clindermatol.2017.06.009
- Reamy BV, Derby R, Bunt CW. Common tongue conditions in primary care. *Am Fam Physician*. 2010;81(5):627-634.
- Gutierrez Gossweiler A, Martinez-Mier EA. Chapter 6: vitamins and oral health. *Monogr Oral Sci*. 2020;28:59-67. doi:10.1159/000455372
- Schlosser BJ, Pirigy M, Mirowski GW. Oral manifestations of hematologic and nutritional diseases. *Otolaryngol Clin North Am*. 2011;44(1):183-vii. doi:10.1016/j.otc.2010.09.007

Lambda Fissuring: Stomach and Pancreatic- Associated Maldigestion



**FM Connection:
Pancreatic exocrine
insufficiency and
hypochlorhydria
associated with
maldigestion-induced
nutrient deficiencies**

Image courtesy of Patrick Hanaway, MD

Tongue Physical Exam Signs: Correlation With Specific Nutrient Deficiencies

Nutrient		Physical Exam Findings
Protein		
Protein Deficiency		Glossitis (atrophic)

Nutrient		Physical Exam Findings
Minerals		
Iron Deficiency		Atrophic glossitis (smooth, glossy, red, oral candidiasis, geographic tongue)
Magnesium Deficiency		Burning tongue (symptoms) – glossopyrosis
Zinc Deficiency		Glossitis, geographic tongue

Tongue Physical Exam Signs: Correlation With Specific Nutrient Deficiencies

Nutrient		Physical Exam Findings
Fat-Soluble Vitamins		
Vitamin A Deficiency		Oral fungal infections (on tongue and mucosa), thickened tongue, geographic tongue
Vitamin D Deficiency		Oral malignancies (mucosal, esophageal, pharyngeal), geographic tongue
Water-Soluble Vitamins		
Riboflavin (B2) Deficiency		Atrophic glossitis (smooth, glossy, red), burning mouth syndrome
Niacin (B3) Deficiency		Atrophic glossitis (smooth, glossy, red), grey pseudomembrane on tongue, swelling and redness of tongue
Vitamin B6 Deficiency		Burning mouth, glossitis, sore tongue, geographic tongue, discoloration of tongue
Folate (B9) Deficiency		Atrophic glossitis (smooth, glossy, red), atrophic papilla, burning mouth syndrome, Hunter's glossitis, oral fungal infections, geographic tongue, oral malignancies (mucosal, esophageal, pharyngeal)
Vitamin B12 Deficiency		Atrophic glossitis (smooth, glossy, red), burning mouth, fissuring of tongue, geographic tongue, Hunter's glossitis, lingual linear lesions
Vitamin C		Oral fungal infections

References

TONGUE PHYSICAL EXAM SIGNS: CORRELATION WITH SPECIFIC NUTRIENT DEFICIENCIES

- Aşkın Ö, Uzunçakmak TKÜ, Altunkalem N, Tüzün Y. Vitamin deficiencies/hypervitaminosis and the skin. *Clin Dermatol*. 2021;39(5):847-857. doi:10.1016/j.clindermatol.2021.05.010
- Bačun B, Galić D, Pul L, Tomas M, Kuiš D. Manifestations and treatment of hypovitaminosis in oral diseases: a systematic review. *Dent J (Basel)*. 2024;12(6):152. doi:10.3390/dj12060152
- Coerdet KM, Goggins CA, Khachemoune A. Vitamins A, B, C, and D: a short review for the dermatologist. *Altern Ther Health Med*. 2021;27(4):41-49.
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Gurvits GE, Tan A. Black hairy tongue syndrome. *World J Gastroenterol*. 2014;20(31):10845-10850. doi:10.3748/wjg.v20.i31.10845
- Gutierrez Gossweiler A, Martinez-Mier EA. Chapter 6: vitamins and oral health. *Monogr Oral Sci*. 2020;28:59-67. doi:10.1159/000455372
- Henkin RI, Gouliouk V, Fordyce A. Distinguishing patients with glossopyrosis from those with oropyrosis based upon clinical differences and differences in saliva and erythrocyte magnesium [published correction appears in *Arch Oral Biol*. 2013;58(1):110]. *Arch Oral Biol*. 2012;57(2):205-210. doi:10.1016/j.archoralbio.2011.08.010
- Lešić S, Ivanišević Z, Špiljak B, Tomas M, Šoštarić M, Včev A. The impact of vitamin deficiencies on oral manifestations in children. *Dent J (Basel)*. 2024;12(4):109. doi:10.3390/dj12040109
- Mulla SA, Ansari A, Bhattacharjee M, Patil AH, Methal SA. Oral manifestations of nutritional deficiencies: a micro review of macro issues in the mouth. *Oral Maxillofac Pathol J*. 2023;14(2):217-219.

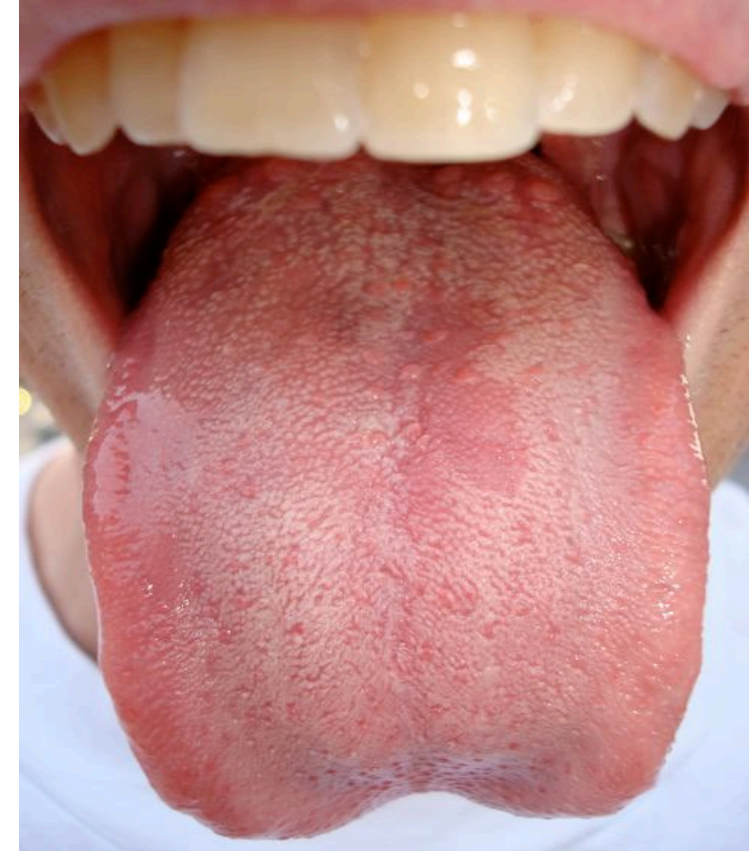
References

TONGUE PHYSICAL EXAM SIGNS: CORRELATION WITH SPECIFIC NUTRIENT DEFICIENCIES

- Radler DR, Lister T. Nutrient deficiencies associated with nutrition-focused physical findings of the oral cavity. *Nutr Clin Pract*. 2013;28(6):710-721. doi:10.1177/0884533613507284
- Rahman N, Walls A. Chapter 12: nutrient deficiencies and oral health. *Monogr Oral Sci*. 2020;28:114-124. doi:10.1159/000455379
- Nagpal B, Nagpal J, Nagpal J, Garg A, Garg A, Nagpal A. Oral manifestations due to nutritional deficiencies & systemic diseases: a review. *Healtalk*. 2019;5.
- Shareef S, Ettefagh L. *Geographic Tongue*. StatPearls Publishing; 2023. Updated July 31, 2023. Accessed June 8, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK554466/>
- Stone PM. *Functional Nutrition Evaluation: The Eight Step Mouth Exam*. The Institute for Functional Medicine; 2014.
- Radler DR, Lister T. Nutrient deficiencies associated with nutrition-focused physical findings of the oral cavity. *Nutr Clin Pract*. 2013;28(6):710-721. doi:10.1177/0884533613507284
- Reamy BV, Derby R, Bunt CW. Common tongue conditions in primary care. *Am Fam Physician*. 2010;81(5):627-634.
- Tolkachjov SN, Bruce AJ. Oral manifestations of nutritional disorders. *Clin Dermatol*. 2017;35(5):441-452. doi:10.1016/j.clindermatol.2017.06.009
- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003

Oral Fungal Infections and Type 2 Diabetes

- In several comparative studies, it was concluded that those diagnosed with T2DM were significantly more likely to also exhibit *Candida albicans* in their saliva:
 - *Candida* was detected in 47% of patients with oral leukoplakia
- Nutrients associated with oral fungal infections: **Vitamins A & C**



Fungal tongue by GrEp is licensed under CC BY-SA 3.0.

- Mohammed MJ, Al-Mizraqchi AS, Ibrahim SM. Oral findings, salivary copper, magnesium, and leptin in type II diabetic patients in relation to oral *Candida* species. *Int J Microbiol.* 2024;2024:8177437. doi:10.1155/2024/8177437
- Jhugroo C, Divakar DD, Jhugroo P, et al. Characterization of oral mucosa lesions and prevalence of yeasts in diabetic patients: a comparative study. *Microb Pathog.* 2019;126:363-367. doi:10.1016/j.micpath.2018.11.028
- See: "References: Tongue Physical Exam Signs: Correlation With Specific Nutrient Deficiencies"

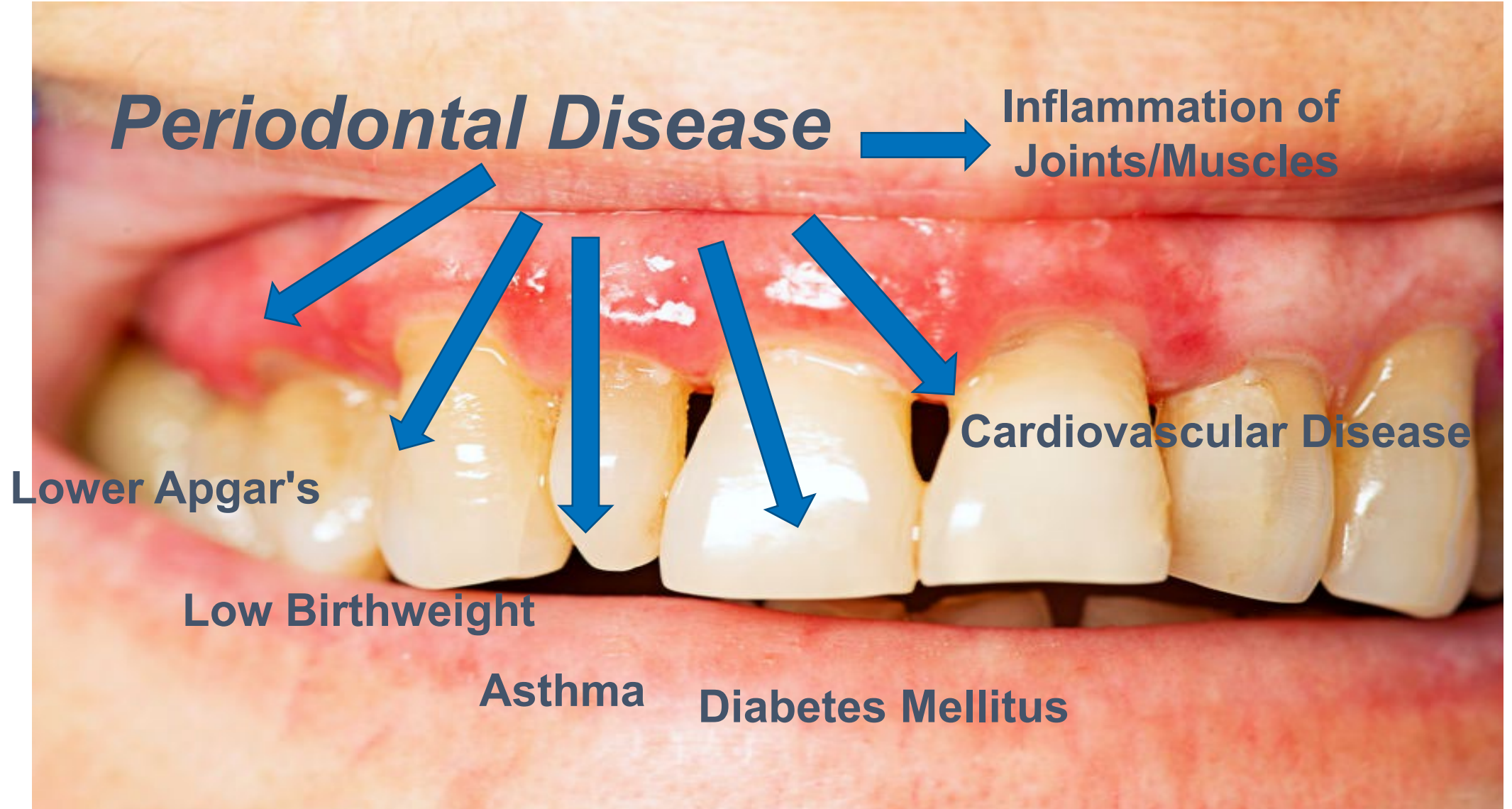
Clinical Focus

- Mouth Exam and Nutrient Associations: Gums and Oral Mucosa



Periodontal Disease

Associated Diseases and Clinical Findings



Gums and Gingiva

Gingivitis



Periodontitis



FM Connection: IL1, IL6, IL10, VDR, genes may be associated with chronic periodontitis and inflammation, driving CVD.

- Neville BW, Damm DD, White DK. *Color Atlas of Clinical Oral Pathology*. 2nd ed. Williams & Wilkins; 2003:100-101.
- Laine ML, Loos BG, Crielaard W. Gene polymorphisms in chronic periodontitis. *Int J Dent*. 2010;2010:324719. doi:10.1155/2010/324719

Periodontal Disease

Associated Nutrient Insufficiencies



FM Connection: Low vitamin D, vitamin C, vitamin A, Zn, Fe, EFAs are all associated with periodontal disease and inflammatory imbalance.

References

PERIODONTAL DISEASE ASSOCIATIONS

- Kim J, Amar S. Periodontal disease and systemic conditions: a bidirectional relationship. *Odontology*. 2006;94(1):10-21. doi:10.1007/s10266-006-0060-6
- Giannobile WV. Host-response therapeutics for periodontal diseases. *J Periodontol*. 2008;79(8 Suppl):1592-1600. doi:10.1902/jop.2008.080174
- Jimenez M, Hu FB, Marino M, Li Y, Joshipura KJ. Prospective associations between measures of adiposity and periodontal disease. *Obesity (Silver Spring)*. 2012;20(8):1718-1725. doi:10.1038/oby.2011.291
- Soory M. Oxidative stress induced mechanisms in the progression of periodontal diseases and cancer: a common approach to redox homeostasis? *Cancers (Basel)*. 2010;2(2):670-692. doi:10.3390/cancers2020670
- Huck O, Tenenbaum H, Davideau JL. Relationship between periodontal diseases and preterm birth: recent epidemiological and biological data. *J Pregnancy*. 2011;2011:164654. doi:10.1155/2011/164654

Minerals Associated With Oral Mucosa Findings

Minerals	Physical Exam Findings
Iron deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, oral candidiasis, oral ulcers, pallor of oral mucosa
Zinc deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, periodontitis, xerostomia (dry mouth)



Fat-Soluble Vitamins Associated With Oral Mucosa Findings

Vitamins	Physical Exam Findings
Vitamin A deficiency	Dry lips, oral fungal infections, gingivitis, periodontitis, salivary gland atrophy/hypofunction, xerostomia (dry mouth)
Vitamin A excess	Gingivitis
Vitamin D deficiency	Aphthous stomatitis (canker sores), malignancies (oral, esophageal, pharyngeal), periodontitis, xerostomia (dry mouth)
Vitamin K deficiency	Ecchymosis of buccal mucosa and palate



See: "References: Oral Mucosa Findings"

B Vitamins Associated With Oral Mucosa Findings

Vitamins	Physical Exam Findings
Thiamine (B1) deficiency	Aphthous stomatitis (canker sores), cracked lips, xerostomia
Riboflavin (B2) deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, pharyngeal edema, pharyngitis
Niacin (B3) deficiency	Gingivitis, oral ulcers, xerostomia (dry mouth)
Pyridoxine (B6) deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, gingivitis, oral ulcers, periodontal disease, stomatitis (angular)
Folate deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, oral candidiasis, gingivitis, stomatitis (angular)
Vitamin B12 deficiency	Aphthous stomatitis (canker sores), hemorrhagic gingivitis, oral ulcers, xerostomia (dry mouth)
Vitamin C deficiency	Black (necrotic) gingiva, oral fungal infections, gum hypertrophy, hemorrhagic gingivitis, periodontal disease, xerostomia (dry mouth)

References

ORAL MUCOSA FINDINGS

- Aşkın Ö, Uzunçakmak TKÜ, Altunkalem N, Tüzün Y. Vitamin deficiencies/hypervitaminosis and the skin. *Clin Dermatol*. 2021;39(5):847-857. doi:10.1016/j.clindermatol.2021.05.010
- Capone K, Sentongo T. The ABCs of nutrient deficiencies and toxicities. *Pediatr Ann*. 2019;48(11):e434-e440. doi:10.3928/19382359-20191015-01
- Cazzolla AP, Cosola MD, Ballini A, et al. The association between nutritional alterations and oral lesions in a pediatric population: an epidemiological study. *Biomed Res Int*. 2021;2021:9992451. doi:10.1155/2021/9992451
- Coerdts KM, Goggins CA, Khachemoune A. Vitamins A, B, C, and D: a short review for the dermatologist. *Altern Ther Health Med*. 2021;27(4):41-49.
- Dipasquale V, Cucinotta U, Romano C. Acute malnutrition in children: pathophysiology, clinical effects and treatment. *Nutrients*. 2020;12(8):2413. doi:10.3390/nu12082413
- Gutierrez Gossweiler A, Martinez-Mier EA. Chapter 6: vitamins and oral health. *Monogr Oral Sci*. 2020;28:59-67. doi:10.1159/000455372
- Kaur K, Sculley D, Wallace J, et al. Micronutrients and bioactive compounds in oral inflammatory diseases. *J Nutr Intermed Metab*. 2019;18:100105. doi:10.1016/j.jnim.2019.100105
- Lešić S, Ivanišević Z, Špiljak B, Tomas M, Šoštarić M, Včev A. The impact of vitamin deficiencies on oral manifestations in children. *Dent J (Basel)*. 2024;12(4):109. doi:10.3390/dj12040109
- Mulla SA, Ansari A, Bhattacharjee M, Patil AH, Methal SA. Oral manifestations of nutritional deficiencies: a micro review of macro issues in the mouth. *Oral Maxillofac Pathol J*. 2023;14(2):217-219.

References

ORAL MUCOSA FINDINGS

- Radler DR, Lister T. Nutrient deficiencies associated with nutrition-focused physical findings of the oral cavity. *Nutr Clin Pract*. 2013;28(6):710-721. doi:10.1177/0884533613507284
- Rahman N, Walls A. Chapter 12: nutrient deficiencies and oral health. *Monogr Oral Sci*. 2020;28:114-124. doi:10.1159/000455379
- Sánchez-Bernal J, Conejero C, Conejero R. Recurrent aphthous stomatitis. Aftosis oral recidivante. *Actas Dermosifiliogr (Engl Ed)*. 2020;111(6):471-480. doi:10.1016/j.ad.2019.09.004
- Stoopler ET, Villa A, Bindakhil M, Díaz DLO, Sollecito TP. Common oral conditions: a review. *JAMA*. 2024;331(12):1045-1054. doi:10.1001/jama.2024.0953
- Strączek A, Szałkowska J, Sutkowska P, et al. Impact of nutrition on the condition of the oral mucosa and periodontium: a narrative review. *Dent Med Probl*. 2023;60(4):697-707. doi:10.17219/dmp/156466
- Tolkachjov SN, Bruce AJ. Oral manifestations of nutritional disorders. *Clin Dermatol*. 2017;35(5):441-452. doi:10.1016/j.clindermatol.2017.06.009
- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003

Clinical Focus

- Mouth Exam and Nutrient Associations: Teeth



Macronutrients Associated With Teeth Findings

Protein		Physical Exam Findings
Protein deficiency		Caries (cavities), enamel hypoplasia, delayed exfoliation of primary teeth, poor structural integrity of dentition
Fats & Oils		Physical Exam Findings
Lipid excess		Caries (cavities)
Carbohydrates		Physical Exam Findings
Carbohydrate excess		Caries (cavities)

References

TEETH FINDINGS

- Aşkın Ö, Uzunçakmak TKÜ, Altunkalem N, Tüzün Y. Vitamin deficiencies/hypervitaminosis and the skin. *Clin Dermatol*. 2021;39(5):847-857. doi:10.1016/j.clindermatol.2021.05.010
- Cazzolla AP, Cosola MD, Ballini A, et al. The association between nutritional alterations and oral lesions in a pediatric population: an epidemiological study. *Biomed Res Int*. 2021;2021:9992451. doi:10.1155/2021/9992451
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Hung M, Patel H, Lee S, Nguyen J, Mohajeri A. The influence of vitamin D levels on dental caries: a retrospective study of the United States population. *Nutrients*. 2024;16(11):1572. doi:10.3390/nu16111572
- Lešić S, Ivanišević Z, Špiljak B, Tomas M, Šoštarić M, Včev A. The impact of vitamin deficiencies on oral manifestations in children. *Dent J (Basel)*. 2024;12(4):109. doi:10.3390/dj12040109
- Mulla SA, Ansari A, Bhattacharjee M, Patil AH, Methal SA. Oral manifestations of nutritional deficiencies: a micro review of macro issues in the mouth. *Oral Maxillofac Pathol J*. 2023;14(2):217-219.
- Shenkin A, Talwar D, Ben-Hamouda N, et al. LLL 44-1 micronutrients in clinical nutrition: trace elements. *Clin Nutr ESPEN*. 2024;61:369-376. doi:10.1016/j.clnesp.2024.04.007
- Strączek A, Szałkowska J, Sutkowska P, et al. Impact of nutrition on the condition of the oral mucosa and periodontium: a narrative review. *Dent Med Probl*. 2023;60(4):697-707. doi:10.17219/dmp/156466
- Tolkachjov SN, Bruce AJ. Oral manifestations of nutritional disorders. *Clin Dermatol*. 2017;35(5):441-452. doi:10.1016/j.clindermatol.2017.06.009

Tooth Enamel Demineralization & Autoimmune Disease: Celiac Disease & Associated Nutrient Insufficiencies



Image courtesy of Michael Stone, MD

Nutrients Associated With Celiac Disease:

- Iodine
- Iron
- Folate
- Vitamin B12
- Vitamin D
- Calcium
- Zinc
- Magnesium

Multiple studies have concluded that there is a strong relationship with dental enamel defects and celiac disease, which may help with early diagnosis, especially in children.

References

NUTRIENTS ASSOCIATED WITH CELIAC DISEASE

- Wieser H, Amato M, Caggiano M, Ciacci C. Dental manifestations and celiac disease—an overview. *J Clin Med*. 2023;12(8):2801. doi:10.3390/jcm12082801
- Verma AK. Nutritional deficiencies in celiac disease: current perspectives. *Nutrients*. 2021;13(12):4476. doi:10.3390/nu13124476
- Caruso R, Pallone F, Stasi E, Romeo S, Monteleone G. Appropriate nutrient supplementation in celiac disease. *Ann Med*. 2013;45(8):522-531. doi:10.3109/07853890.2013.849383
- Rubio-Tapia A, Hill ID, Semrad C, et al. American College of Gastroenterology Guidelines update: diagnosis and management of celiac disease [published correction appears in *Am J Gastroenterol*. 2024;119(7):1441. doi:10.14309/ajg.0000000000002210]. *Am J Gastroenterol*. 2023;118(1):59-76. doi:10.14309/ajg.0000000000002075
- Kreutz JM, Adriaanse MPM, van der Ploeg EMC, Vreugdenhil ACE. Narrative review: nutrient deficiencies in adults and children with treated and untreated celiac disease. *Nutrients*. 2020;12(2):500. doi:10.3390/nu12020500

Micronutrients Associated With Teeth Findings

Minerals	Physical Exam Findings
Calcium deficiency	Missing teeth
Zinc deficiency	Caries (cavities), missing teeth

Vitamins	Physical Exam Findings
Vitamin A deficiency	Caries (cavities), enamel hypoplasia, poor structural integrity of dentition, tooth morphogenesis defects
Vitamin D deficiency	Caries (cavities), enamel hypoplasia, enamel hypomineralization
Thiamine (B1) deficiency	Enamel hypomineralization
Riboflavin (B2) deficiency	Enamel hypomineralization
Niacin (B3) deficiency	Caries (cavities)
Vitamin C deficiency	Missing teeth

See: "References: Teeth Findings"

Oral Health Exam

Missing Teeth



Hypomineralization



Procaccini M, Campisi G, Bufo P, et al. Lack of association between celiac disease and dental enamel hypoplasia in a case-control study from an Italian central region. *Head Face Med.* 2007;3:25. doi:10.1186/1746-160X-3-25
<https://creativecommons.org/licenses/by/2.0/>

Fluorosis



Mild fluorosis



Moderate fluorosis



Severe fluorosis

**Most common mediator:
fluoridated drinking water**

- Images: Fluorosis. (2019, March 08). Retrieved April 16, 2021, from https://www.cdc.gov/fluoridation/faqs/dental_fluorosis/index.htm
- Centers for Disease Control and Prevention. Community water fluoridation: frequently asked questions. Updated May 15, 2024. Accessed June 8, 2025. <https://www.cdc.gov/fluoridation/faq/index.html>
- Graves JM, Daniell W, James F, Milgrom P. Estimating fluoride exposure in rural communities: a case study in Western Washington. *Wash State J Public Health Pract.* 2009;2(2):22-31.
- Hudak PF. Elevated fluoride and selenium in west Texas groundwater. *Bull Environ Contam Toxicol.* 2009;82(1):39-42. doi:10.1007/s00128-008-9583-6
- DenBesten P, Li W. Chronic fluoride toxicity: dental fluorosis. *Monogr Oral Sci.* 2011;22:81-96. doi:10.1159/000327028

Chewing and Swallowing

Indirect associations with nutritional imbalances:

- Missing teeth – ill-fitting dentures, chronic inflammation, or sores
- Hyposalivation, dry mouth
- These factors are likely to deter the patient's selection, digestion, and assimilation of nutrient-dense whole foods



Sense of Smell and Taste

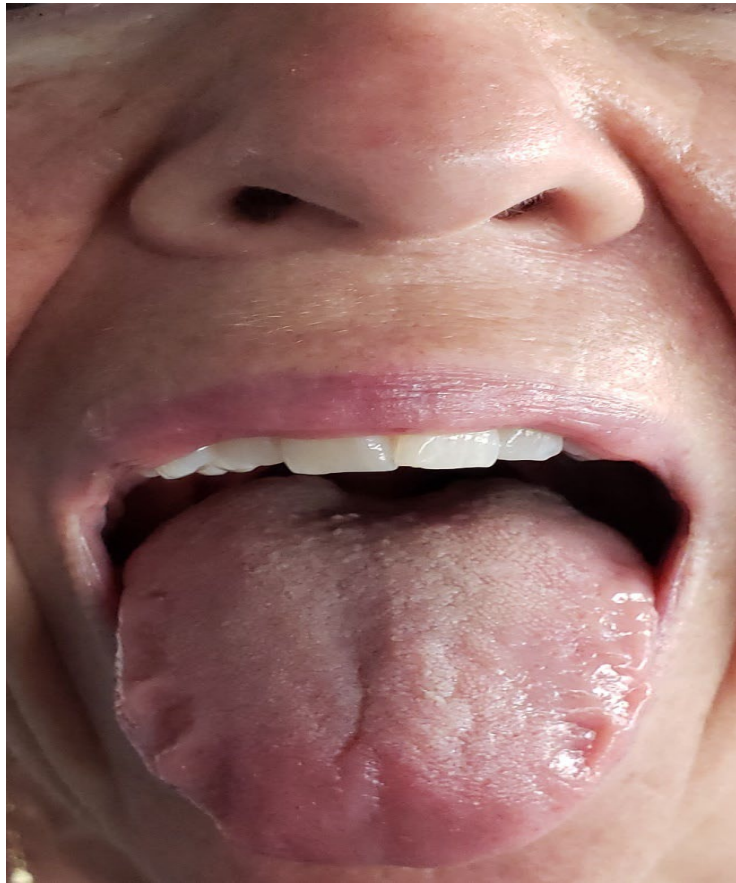


Image courtesy of Michael Stone, MD

- Zinc
- Vitamin A
- B Vitamins
- Vitamin D
- Vitamin E
- Copper
- Iodine
- Iron

Loss of olfaction in old age is a frequent problem, which occurs at the same frequency as diabetes or severe sensorineural hearing loss.

Plays a role in retronasal smell, which accounts for 20% of “smell.”

References

NUTRIENTS ASSOCIATED WITH TASTE AND SMELL

Zinc:

- Mattes RD, Cowart BJ. Dietary assessment of patients with chemosensory disorders. *J Am Diet Assoc.* 1994;94(1):50-56. doi:10.1016/0002-8223(94)92041-9

Vitamins A, D, and B:

- Mulla SA, Ansari A, Bhattacharjee M, Patil AH, Methal SA. Oral manifestations of nutritional deficiencies: a micro review of macro issues in the mouth. *Oral Maxillofac Pathol J.* 2023;14(2):217-219.

B Vitamins:

- Nagpal B, Nagpal J, Nagpal J, Garg A, Garg A, Nagpal A. Oral manifestations due to nutritional deficiencies & systemic diseases: a review. *Healtalk.* 2019;5.
- Tolkachjov SN, Bruce AJ. Oral manifestations of nutritional disorders. *Clin Dermatol.* 2017;35(5):441-452. doi:10.1016/j.clindermatol.2017.06.009
- Coerdts KM, Goggins CA, Khachemoune A. Vitamins A, B, C, and D: a short review for the dermatologist. *Altern Ther Health Med.* 2021;27(4):41-49.

Vitamin E:

- Henkin RI, Hoetker JD. Deficient dietary intake of vitamin E in patients with taste and smell dysfunctions: is vitamin E a cofactor in taste bud and olfactory epithelium apoptosis and in stem cell maturation and development? *Nutrition.* 2003;19(11-12):1013-1021. doi:10.1016/j.nut.2003.08.006

Minerals (Cu, I, Fe):

- Kershaw JC, Mattes RD. Nutrition and taste and smell dysfunction. *World J Otorhinolaryngol Head Neck Surg.* 2018;4(1):3-10. doi:10.1016/j.wjorl.2018.02.006
- Radler DR, Lister T. Nutrient deficiencies associated with nutrition-focused physical findings of the oral cavity. *Nutr Clin Pract.* 2013;28(6):710-721. doi:10.1177/0884533613507284

ABCDs of Nutritional Evaluation

Part C: Clinical Findings-Nutrition Physical Exam

Part 2

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Skin Exam and Nutrient Associations



The ABCDs of Nutrition Evaluation Skin Exam



Skin Exam

Factors to Consider:

- Temperature
- Texture
- Color
- Hydration
- Lesions
- Hair Distribution



Xerosis

Nutrient Deficiencies:

EFA's

Zinc

Vitamin A

Vitamin C

Vitamin B

Hyperkeratosis Pilaris

Images courtesy of Michael Stone, MD

Seborrhea Can Be Subtle



Associated Nutrients:

- Vitamin B6
- Vitamin B2
- Vitamin B12
- Biotin
- Zinc
- Copper

Image courtesy of Michael Stone, MD

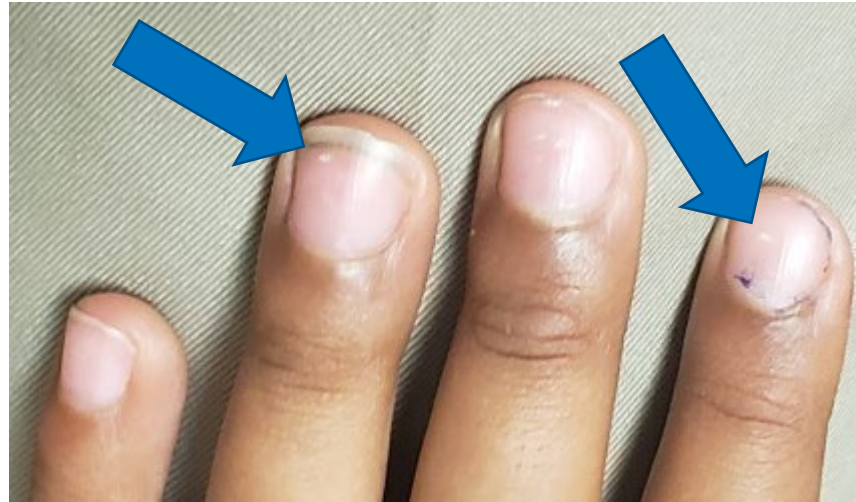
References

SKIN AND NUTRIENTS

- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003
- Gade A, Hwang JR, Hoegler K, Khan S, Khachemoune A. Therapeutic use of trace elements in dermatology. *Altern Ther Health Med*. 2023;29(4):246-252.
- O'Connor K, Goldberg LJ. Nutrition and hair. *Clin Dermatol*. 2021;39(5):809-818. doi:10.1016/j.clindermatol.2021.05.008
- Dipasquale V, Cucinotta U, Romano C. Acute malnutrition in children: pathophysiology, clinical effects and treatment. *Nutrients*. 2020;12(8):2413. doi:10.3390/nu12082413
- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Gade A, Hwang JR, Hoegler K, Khan S, Khachemoune A. Therapeutic use of trace elements in dermatology. *Altern Ther Health Med*. 2023;29(4):246-252.
- Podgórska A, Kicman A, Naliwajko S, Waciewicz-Muczyńska M, Niczyporuk M. Zinc, copper, and iron in selected skin diseases. *Int J Mol Sci*. 2024;25(7):3823. doi:10.3390/ijms25073823
- Dattola A, Silvestri M, Bennardo L, et al. Role of vitamins in skin health: a systematic review. *Curr Nutr Rep*. 2020;9(3):226-235. doi:10.1007/s13668-020-00322-4

Clinical Findings & Nutrient Associations

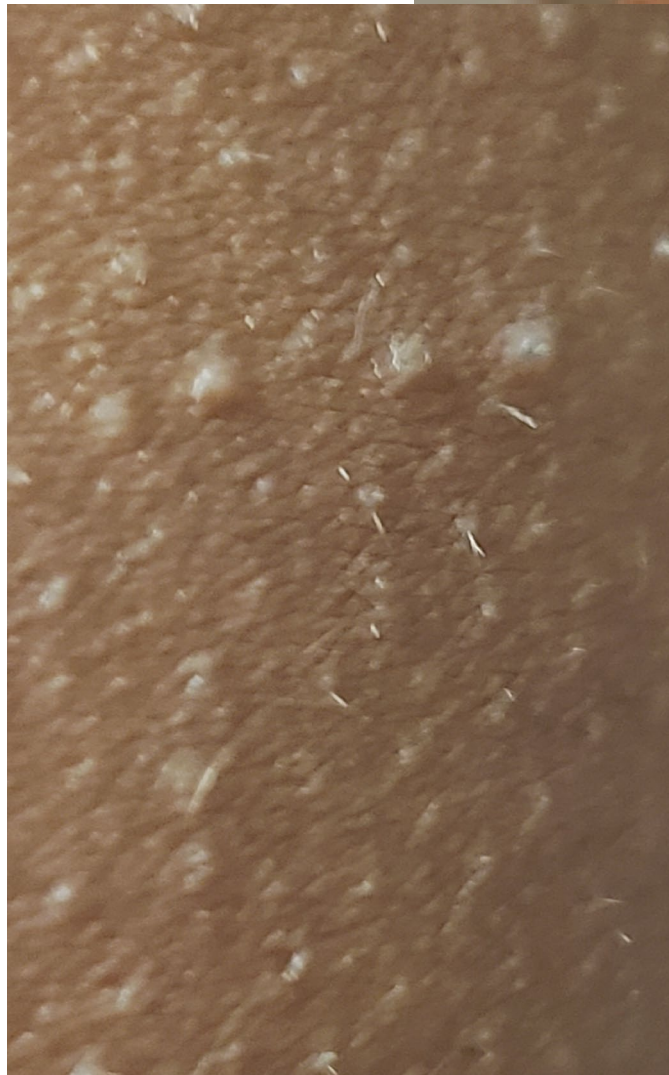
Acne, Leukonychia, Hyperkeratosis Pilaris



Zinc insufficiency common with these three findings
Vitamin A, EFA common with acne and hyperkeratosis pilaris



Hyperkeratosis Pilaris





EFA Deficiency – Palmar xerosis, mild thickening

Image courtesy of Michael Stone, MD

Dermatologic Signs of Essential Fatty Acid Deficiency

- Alopecia: Hair loss
- Eczematous Dermatitis: Presented as dry, scaly, and itchy skin, often resembling eczema
- Generalized Dry, Rough Skin
- Poor Wound Healing
- Periorificial Dermatitis: Dermatitis around the mouth, eyes, and nose
- Acrodermatitis: Dermatitis affecting the extremities, especially hands and feet

These signs suggest the body cannot maintain or repair the skin barrier due to a deficiency in essential fatty acids like linoleic and alpha-linolenic acid, leading to dermatological manifestations.

Follicular Hyperkeratosis: Vitamin A, Essential Fatty Acids

Before Treatment



After Treatment



Images: Murthy SR, Prabhakaran VC. Phrynoderma and night blindness. Indian J Ophthalmol. 2010;58(2):175-176. doi:10.4103/0301-4738.60089. Licensed under CC BY-NC-NA.

Treatment: Vitamin A (100,000 IU IM)

References

FOLLICULAR HYPERKERATOSIS

- Ragunatha S, Jagannath Kumar V, Murugesh SB, Ramesh M, Narendra G, Kapoor M. Therapeutic response of vitamin A, vitamin B complex, essential fatty acids (EFA) and vitamin E in the treatment of phrynoderma: a randomized controlled study. *J Clin Diagn Res*. 2014;8(1):116-118. doi:10.7860/JCDR/2014/7086.3918
- Loewenthal LJA. A new cutaneous manifestation in the syndrome of vitamin A deficiency. *Arch Derm Syphilol*. 1933;28(5):700-708. doi:10.1001/archderm.1933.01460050094008
- Barthelemy H, Chouvet B, Cambazard F. Skin and mucosal manifestations in vitamin deficiency. *J Am Acad Dermatol*. 1986;15(6):1263-1274. doi:10.1016/s0190-9622(86)70301-0
- Maronn M, Allen DM, Esterly NB. Phrynoderma: a manifestation of vitamin A deficiency?... The rest of the story. *Pediatr Dermatol*. 2005;22(1):60-63. doi:10.1111/j.1525-1470.2005.22113.x
- Ragunatha S, Kumar VJ, Murugesh SB. A clinical study of 125 patients with phrynoderma. *Indian J Dermatol*. 2011;56(4):389-392. doi:10.4103/0019-5154.84760
- Ansstas G, Gopalswamy N, Thakore J. Vitamin A deficiency. Medscape. Updated February 3, 2025. Accessed June 8, 2025. <https://emedicine.medscape.com/article/126004-overview>
- Murthy SR, Prabhakaran VC. Phrynoderma and night blindness. *Indian J Ophthalmol*. 2010;58(2):175-176. doi:10.4103/0301-4738.60089

Dry Skin Rashes



Seborrheic dermatitis by Roymishali is licensed under CC BY-SA 3.0.
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Images courtesy of Michael Stone, MD

Seborrhea

**Nutrient Deficiencies:
EFA, Zinc, Vitamin A, Biotin**

- McLaren DS. *A Colour Atlas and Text of Diet-Related Disorders*. 2nd ed. Wolfe Pub; 1992.
- Calder PC, Kremmyda LS, Vlachava M, Noakes PS, Miles EA. Is there a role for fatty acids in early life programming of the immune system? *Proc Nutr Soc*. 2010;69(3):373-380. doi:10.1017/S0029665110001552
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Images courtesy of Michael Stone, MD

**Nutrient Deficiencies:
EFA, Vitamin A, Zinc**



Image courtesy 123RF



Eczema

Clinical Findings & Nutrient Connections: Psoriasis



- GI Barrier Integrity
- Vitamin A
- Essential Fatty Acids
- Vitamin D
- Bile Insufficiency
- Dietary Preferences



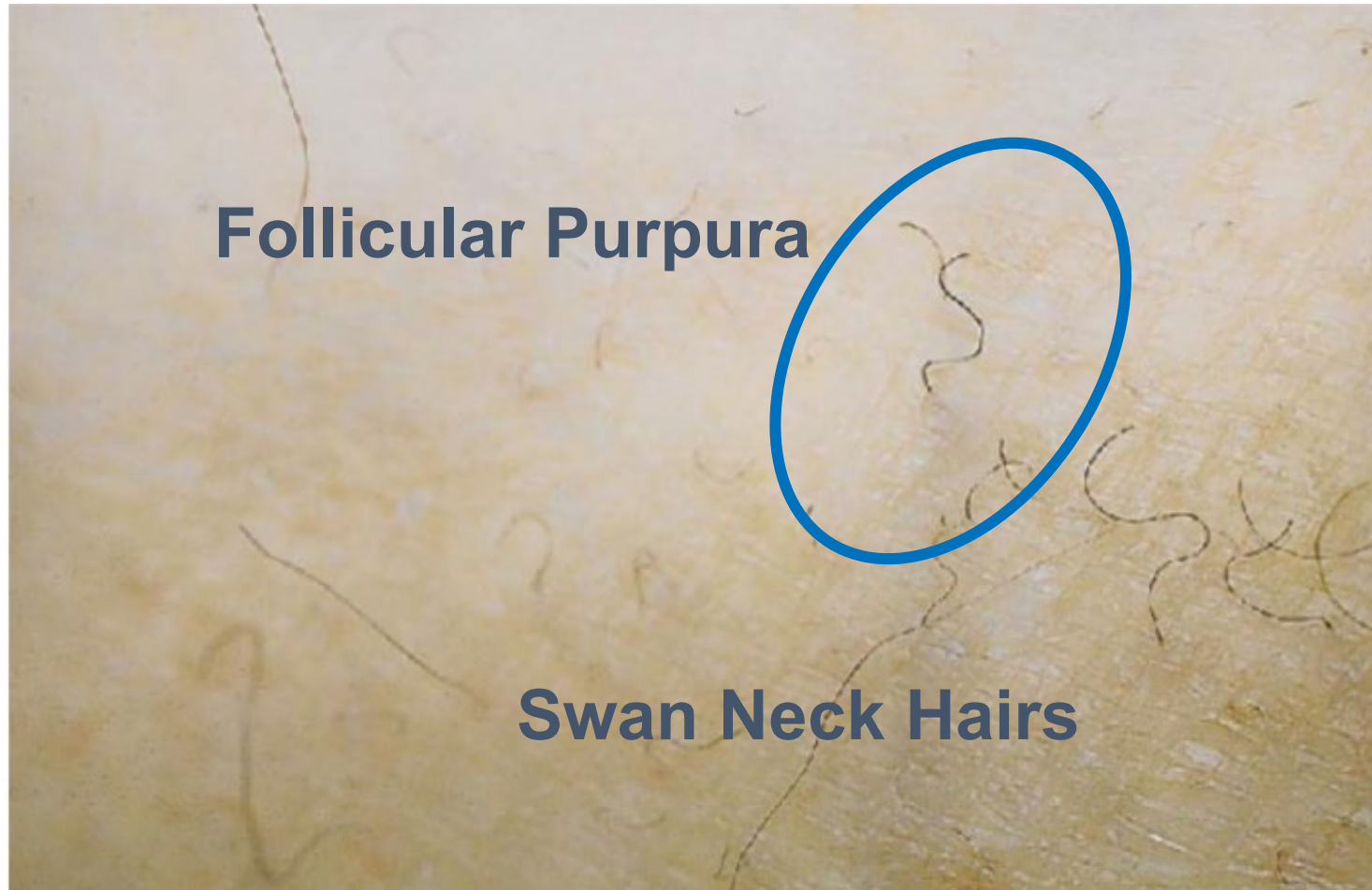
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References

PSORIASIS

- Polak K, Bergler-Czop B, Szczepanek M, Wojciechowska K, Frątczak A, Kiss N. Psoriasis and gut microbiome-current state of art. *Int J Mol Sci*. 2021;22(9):4529. doi:10.3390/ijms22094529
- Kapoor B, Gulati M, Rani P, Gupta R. Psoriasis: interplay between dysbiosis and host immune system. *Autoimmun Rev*. 2022;21(11):103169. doi:10.1016/j.autrev.2022.103169
- Kanda N, Hoashi T, Saeki H. Nutrition and psoriasis. *Int J Mol Sci*. 2020;21(15):5405. doi:10.3390/ijms21155405
- Jimenez-Sanchez M, Celiberto LS, Yang H, Sham HP, Vallance BA. The gut-skin axis: a bi-directional, microbiota-driven relationship with therapeutic potential. *Gut Microbes*. 2025;17(1):2473524. doi:10.1080/19490976.2025.2473524
- Zhu Q, Wu K, Yang Q, Meng B, Niu Y, Zhao F. Advances in psoriasis and gut microorganisms with co-metabolites. *Front Microbiol*. 2023;14:1192543. doi:10.3389/fmicb.2023.1192543
- Yegorov S, Babenko D, Kozhakhmetov S, et al. Psoriasis is associated with elevated gut IL-1 α and intestinal microbiome alterations. *Front Immunol*. 2020;11:571319. doi:10.3389/fimmu.2020.571319
- Cintoni M, Palombaro M, Maramao FS, et al. Metabolic disorders and psoriasis: exploring the role of nutritional interventions. *Nutrients*. 2023;15(18):3876. doi:10.3390/nu15183876

Vitamin C Deficiency



Bruising



- Reprinted from: Hirschmann JV, Raugi GJ. Adult scurvy. *J Am Acad Dermatol*. 1999;41(6):895-910. doi:10.1016/s0190-9622(99)70244-6. Copyright 1999, with permission from American Academy of Dermatology, Inc.
- Krečak I, Babić G, Skelin M. Scurvy. *Acta Dermatovenerol Croat*. 2022;30(1):59-60.

Perifollicular Petechiae

VITAMIN C OR VITAMIN K DEFICIENCY
(also bioflavonoid insufficiency)



Image courtesy of the CDC

Zinc Deficiency Rash – Acrodermatitis Enteropathica

Clinical Triad:

- Acral dermatitis
- Alopecia
- Diarrhea

Potential Biomarkers:

- Low alkaline phosphatase
- Low serum zinc



Acne

Associated Nutrient Insufficiencies:

- EFA
- Zinc
- Selenium
- Vitamins A, C, E,
- Vitamin B6
- Pantothenic Acid



- Acne vulgaris on an 18-year-old male by Roshu Bangal is licensed under [CC BY-SA 4.0](#).
- Cystic acne on back and neck by James Heilman, MD, is licensed under [CC BY-SA 3.0](#).

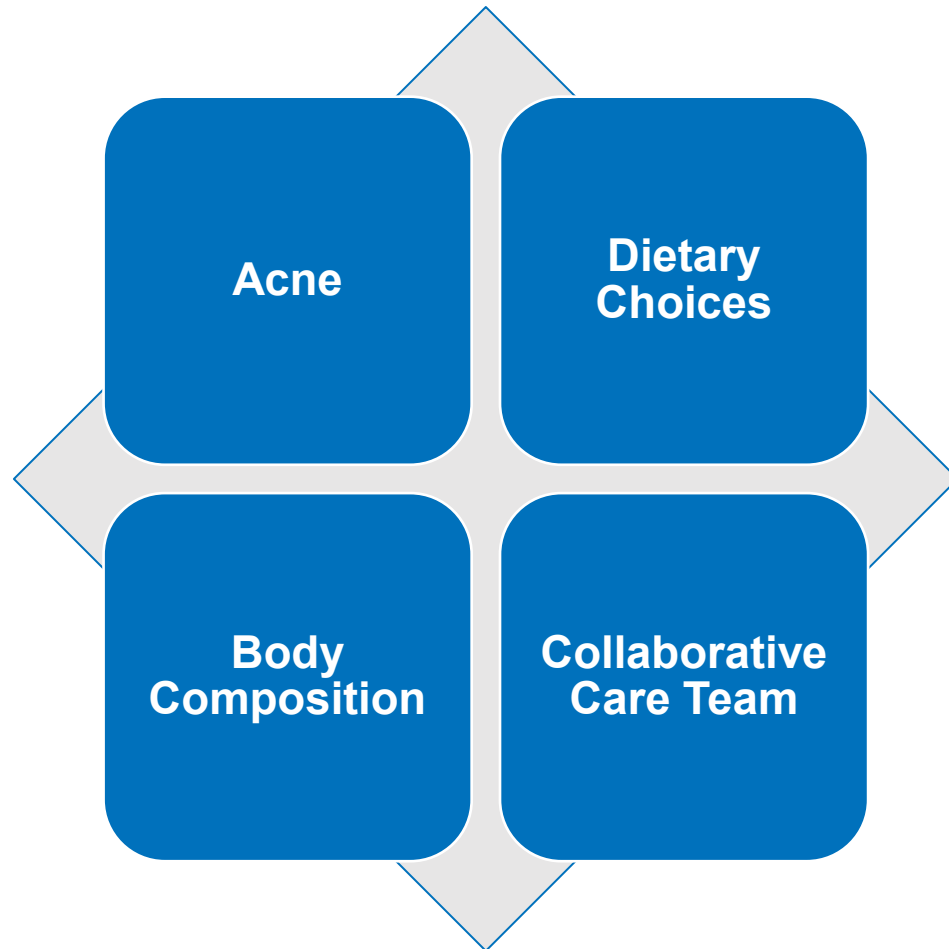
References

ACNE

- Tunçez Akyürek F, Saylam Kurtipek G, Kurku H, et al. Assessment of ADMA, IMA, and vitamin A and E levels in patients with acne vulgaris. *J Cosmet Dermatol*. 2020;19(12):3408-3413. doi:10.1111/jocd.13590
- Popa GL, Mitran CI, Mitran MI, et al. Markers of oxidative stress in patients with acne: a literature review. *Life (Basel)*. 2023;13(7):1433. doi:10.3390/life13071433
- Liu X, Yang G, Luo M, et al. Serum vitamin E levels and chronic inflammatory skin diseases: a systematic review and meta-analysis. *PLoS One*. 2021;16(12):e0261259. doi:10.1371/journal.pone.0261259
- Lv J, Ai P, Lei S, Zhou F, Chen S, Zhang Y. Selenium levels and skin diseases: systematic review and meta-analysis. *J Trace Elem Med Biol*. 2020;62:126548. doi:10.1016/j.jtemb.2020.126548
- Podgórska A, Kicman A, Naliwajko S, Wacewicz-Muczyńska M, Niczyporuk M. Zinc, copper, and iron in selected skin diseases. *Int J Mol Sci*. 2024;25(7):3823. doi:10.3390/ijms25073823

Clinical Findings & Nutrient Connections:

Acne, Diet, Body Composition, Phase Angle (BIA)



- Patients with acne showed poorer body composition, lower phase angle, and reduced adherence to the Mediterranean diet.
- More severe acne, based on the Global Acne Grading Scale and Predimed score, was associated with a phase angle below 6.1, indicating lower compliance with the anti-inflammatory diet.

Conclusions: Collaboration with dermatologists and nutritionists may provide an effective collaborative care model for the complex management of acne.

Macronutrients Associated With Skin Findings

Protein	Physical Exam Findings
Protein deficiency	Dry, loose, shiny, wrinkled skin; erythematous pigmentation, exfoliative erythroderma, swelling (edema) of extremities

Fats & Oils	Physical Exam Findings
EFA deficiency	Dermatitis (seborrheic-like), desquamation, phrynoderma (toad skin), poor wound healing, scaly and dry rash

Vitamins Associated With Skin Findings

Vitamins		Physical Exam Findings
Fat-Soluble		
Vitamin A deficiency		Follicular hyperkeratosis, phrynoderma (toad skin), xerosis
Vitamin A excess		Cheilitis, dermatitis (seborrheic-like), desquamation, pruritic, xerosis
Vitamin D deficiency		Atopic dermatitis, morphea, polymorphous light eruption, psoriasis, skin cancer
Vitamin E deficiency		Phrynoderma (toad skin), xerosis
Vitamin K deficiency		Petechiae, purpura

Phytonutrients		Physical Exam Findings
Carotenoid excess		Carotenoderma (yellow-orange discoloration)

Vitamins Associated With Skin Findings

Vitamins Physical Exam Findings	
Water-Soluble	
Riboflavin (B2) deficiency	Angular stomatitis, cheilitis, dermatitis (generalized, atopic, or seborrheic)
Niacin (B3) deficiency	Casal's necklace, cheilitis, hyperkeratosis, hyperpigmentation, photo-distributed dermatitis
Vitamin B6	Cheilitis, glossitis, periorificial dermatitis, photo-distributed dermatitis, seborrheic dermatitis
Biotin (B7)	Dermatitis (eczematous, seborrheic)
Folate (B9)	Angular cheilitis, hyperpigmentation
Vitamin B12	Aphthous stomatitis, angular cheilitis, depigmentation, glossitis, hyperpigmentation, seborrheic dermatitis
Vitamin C	Easy bleeding/bruising, edema (lower extremity), follicular hyperkeratosis, petechiae/purpura, phrynoderma (toad skin), poor wound healing

Minerals Associated With Skin Findings

Minerals	Physical Exam Findings
Copper deficiency	Depigmentation
Copper excess	Psoriasis, seborrheic dermatitis
Iron deficiency	Angular stomatitis, cheilitis (angular), pallor, pruritis, psoriasis
Iron excess	Hyperpigmentation
Manganese deficiency	Dermatitis (seborrheic-like)
Selenium deficiency	Acne vulgaris, epidermolysis bullosa (fragile, blistering skin), depigmentation, psoriasis
Zinc deficiency	Acne, acrodermatitis enteropathica, cheilitis, dermatitis (atopic, diaper, seborrheic) poor wound healing, psoriasis, seborrheic dermatitis, xerosis

References

SKIN FINDINGS AND THEIR NUTRIENT ASSOCIATIONS

- Aşkın Ö, Uzunçakmak TKÜ, Altunkalem N, Tüzün Y. Vitamin deficiencies/hypervitaminosis and the skin. *Clin Dermatol*. 2021;39(5):847-857. doi:10.1016/j.clindermatol.2021.05.010
- Capone K, Sentongo T. The ABCs of nutrient deficiencies and toxicities. *Pediatr Ann*. 2019;48(11):e434-e440. doi:10.3928/19382359-20191015-01
- Coerdts KM, Goggins CA, Khachemoune A. Vitamins A, B, C, and D: a short review for the dermatologist. *Altern Ther Health Med*. 2021;27(4):41-49.
- Dattola A, Silvestri M, Bennardo L, et al. Role of vitamins in skin health: a systematic review. *Curr Nutr Rep*. 2020;9(3):226-235. doi:10.1007/s13668-020-00322-4
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Dipasquale V, Cucinotta U, Romano C. Acute malnutrition in children: pathophysiology, clinical effects and treatment. *Nutrients*. 2020;12(8):2413. doi:10.3390/nu12082413
- Gade A, Hwang JR, Hoegler K, Khan S, Khachemoune A. Therapeutic use of trace elements in dermatology. *Altern Ther Health Med*. 2023;29(4):246-252.
- Kiani AK, Dhuli K, Donato K, et al. Main nutritional deficiencies. *J Prev Med Hyg*. 2022;63(2 Suppl 3):E93-E101. doi:10.15167/2421-4248/jpmh2022.63.2S3.2752
- Lv J, Ai P, Lei S, Zhou F, Chen S, Zhang Y. Selenium levels and skin diseases: systematic review and meta-analysis. *J Trace Elem Med Biol*. 2020;62:126548. doi:10.1016/j.jtemb.2020.126548
- O'Connor K, Goldberg LJ. Nutrition and hair. *Clin Dermatol*. 2021;39(5):809-818. doi:10.1016/j.clindermatol.2021.05.008
- Podgórska A, Kicman A, Naliwajko S, Waciewicz-Muczyńska M, Niczyporuk M. Zinc, copper, and iron in selected skin diseases. *Int J Mol Sci*. 2024;25(7):3823. doi:10.3390/ijms25073823
- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003

ABCDs of Nutritional Evaluation

Part C: Clinical Findings-Nutrition Physical Exam

Part 3

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Nail Exam and Nutrient Associations



The ABCDs of Nutrition Evaluation Nail Exam



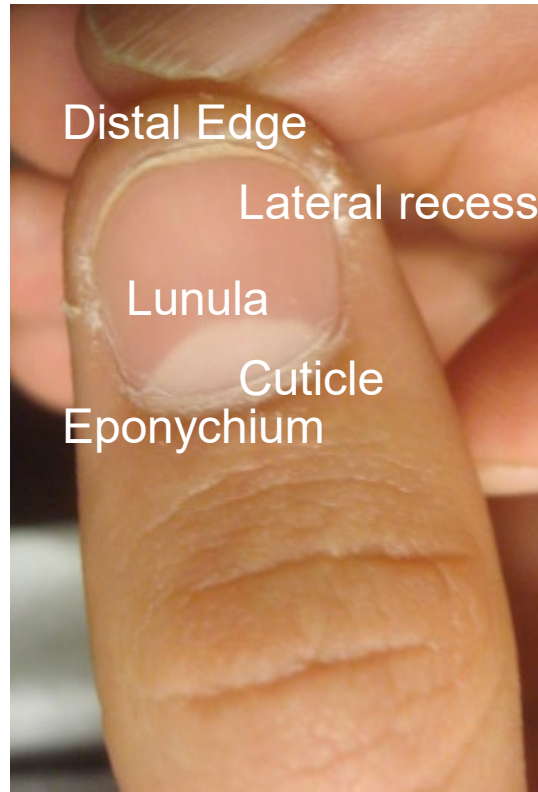
Nails are the “tree rings”
of our physiology

Six months of your
patient’s medical record



Nail Exam

Factors to Consider and Normal Findings



- Shape
- Color
- Pattern of Color
- Texture and Strength
- Growth Pattern
- Surrounding Tissue

Images courtesy of Michael Stone, MD

Normal = smooth, translucent, slightly curved nail surface and firmly attached to nail bed; nail beds have a capillary refill <3 seconds.

Koilonychia = Spooned Nail

Insufficiencies: Protein, Iron, Chromium, Copper, Zinc



FM Connection: Patients with iron deficiency have greater issues with immune dysregulation, increased inflammation, endothelial dysfunction, ASCVD plaque, greater zinc insufficiency, aneurysms associated with low copper.

- "Spooning of the Nails" by CHEITZ is licensed under [CC BY 2.0](#).
- Razmi TM, De D. Bead retention test in koilonychia. *Indian J Dermatol Venereol Leprol*. 2019;85(2):229-230. doi:10.4103/ijdv.IJDVL_350_18. Published by [Scientific Scholar](#) on behalf of Indian Association of Dermatologists, Venereologists & Leprologists (IADVL).

Leukonychia Punctata & Leukonychia Striata

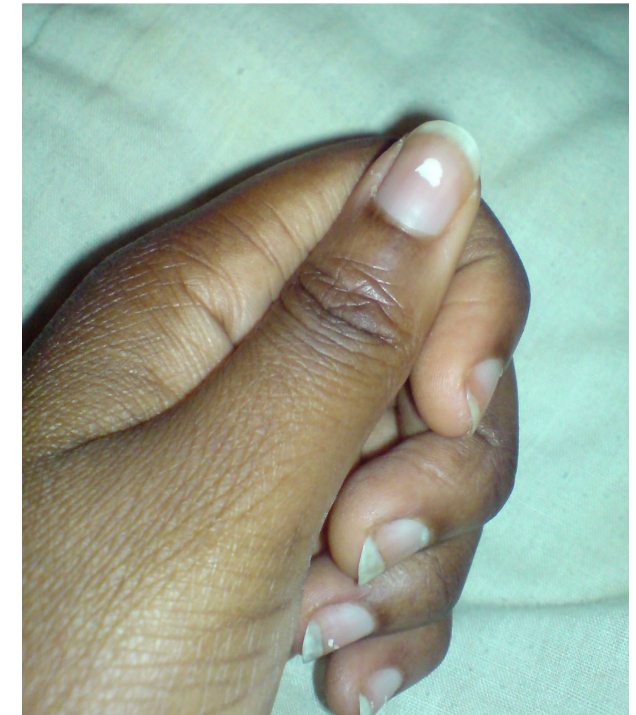
Zinc & Selenium Deficiency



Image courtesy of Michael Stone, MD



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Weinberg's Color Atlas of Pediatric Dermatology. 5th ed. McGraw
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Clearance Center, Inc.



Miles DW, Rubens RD. Images in clinical medicine.
Transverse leukonychia. *N Engl J Med*. 1995;333(2):100.
doi:10.1056/NEJM199507133330205
Credit: Londonsista.

Thin, Brittle, Cracking Nails

FM Connection:

- Nutrients: Iron, Vitamin A, Vitamin C, B6, Protein, EFAs, Biotin, Zinc and Calcium Deficiency
- Other: Thyroid, Osteoporosis



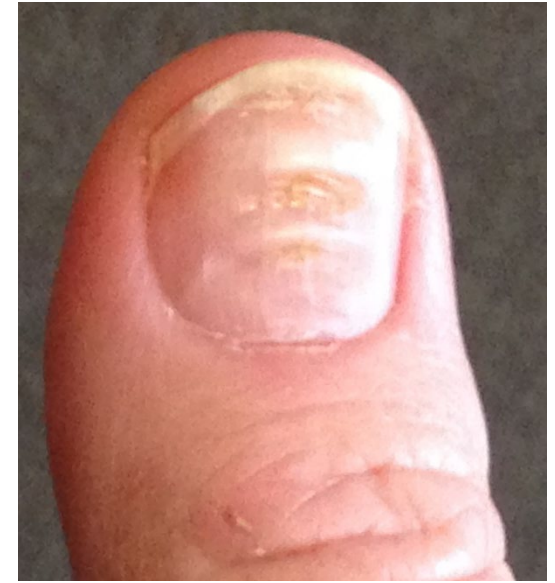
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Texture and Strength, Growth Pattern

Beau's Lines

- Zinc, protein, niacin deficits
- Other causes:
 - Severe illness
 - Measles/mumps
 - Syphilis
 - Poorly controlled DM
 - Myocarditis

Growth arrest
2-3 months ago



Beau's Lines by LynnMcCleary is licensed under CC BY-SA 4.0.



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Clinical Findings

Eggshell-Thin Nails With Peeling and Paronychia



Image courtesy of Michael Stone, MD

Mixed Clinical Assessment:

- EFA Deficiency
- Vitamin D Deficiency
- Calcium Insufficiency
- Iron Insufficiency

Nutrient-Associated Nail Findings

Protein	Physical Exam Findings
Protein deficiency	Beau's lines, brittle nails, hapalonychia (eggshell nails), koilonychia (spoon-shaped nails; especially in deficiencies of sulfur-containing amino acids: cysteine or methionine)

Minerals	Physical Exam Findings
Calcium deficiency	Beau's lines, brittle nails, hapalonychia (eggshell nails), longitudinal striations/ridges (onychorrhexis), onychomadesis, Mee's lines (transverse leukonychia)
Copper excess	Azure (blue) lunula
Iron deficiency	Brittle nails, central ridging, koilonychia (spoon-shaped nails), longitudinal striations/ridges (onychorrhexis), onycholysis, pallor of nails
Iron excess	Melanonychia (brown or black coloring), splinter hemorrhages of nail bed, Mee's lines (transverse leukonychia)
Magnesium deficiency	Brittle nails
Selenium	Brittle nails, pallor of nails, leukonychia (white nails)
Zinc	Beau's lines, brittle nails, longitudinal striations/ridges (onychorrhexis), Muehrcke's lines, nail dystrophy, paronychia, Mee's lines (transverse leukonychia)

See: "References: Nail Findings and Their Nutrient Associations"

Nutrient-Associated Nail Findings

Vitamins	Physical Exam Findings
Vitamin A deficiency	Brittle nails, hapalonychia (eggshell nails)
Vitamin A excess	Clubbing
Vitamin D deficiency	Brittle nails, melanonychia (brown or black coloring), hapalonychia (eggshell nails)
Vitamin E	Brittle nails
Riboflavin (B2)	Koilonychia (spoon-shaped nails)
Niacin (B3)	Beau's lines, koilonychia (spoon-shaped nails), longitudinal hyperpigmented streaks, onycholysis, Mee's lines (transverse leukonychia)
Vitamin B6	Hapalonychia (eggshell nails)
Biotin (B7)	Brittle nails
Vitamin B12	Azure (blue) lunula, longitudinal hyperpigmented streaks, melanonychia (brown or black coloring)
Vitamin C	Brittle nails, hapalonychia (eggshell nails), koilonychia (spoon-shaped nails), splinter hemorrhages of nail bed

See: "References: Nail Findings and Their Nutrient Associations"

References

NAIL FINDINGS AND THEIR NUTRIENT ASSOCIATIONS

- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Fawcett RS, Linford S, Stulberg DL. Nail abnormalities: clues to systemic disease. *Am Fam Physician*. 2004;69(6):1417-1424.
- Finner AM. Nutrition and hair: deficiencies and supplements. *Dermatol Clin*. 2013;31(1):167-172. doi:10.1016/j.det.2012.08.015
- Gade A, Hwang JR, Hoegler K, Khan S, Khachemoune A. Therapeutic use of trace elements in dermatology. *Altern Ther Health Med*. 2023;29(4):246-252.
- Iorizzo M, Starace M, Pasch MC. Leukonychia: what can white nails tell us? *Am J Clin Dermatol*. 2022;23(2):177-193. doi:10.1007/s40257-022-00671-6
- Lee DK, Lipner SR. Optimal diagnosis and management of common nail disorders. *Ann Med*. 2022;54(1):694-712. doi:10.1080/07853890.2022.2044511
- Muddasani S, Lin G, Hooper J, Sloan SB. Nutrition and nail disease. *Clin Dermatol*. 2021;39(5):819-828. doi:10.1016/j.clindermatol.2021.05.009
- Seshadri D, De D. Nails in nutritional deficiencies. *Indian J Dermatol Venereol Leprol*. 2012;78(3):237-241. doi:10.4103/0378-6323.95437
- Williams ME. Examining the fingernails when evaluating presenting symptoms in elderly patients. Medscape Family Medicine. Published November 23, 2009. Accessed May 29, 2025. http://www.medscape.com/viewarticle/712251_1
- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003

ABCDs of Nutritional Evaluation

Part C: Clinical Findings-Nutrition Physical Exam

Part 4

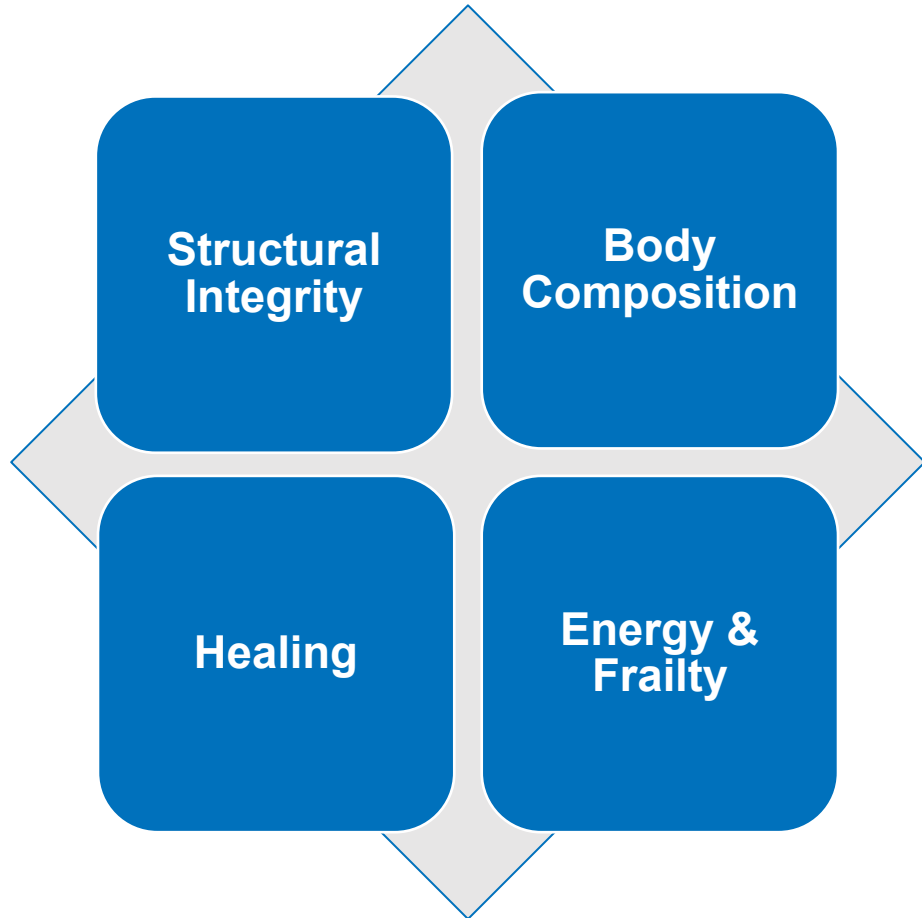
APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Practical application: common patterns of nutrition exam findings by nutrient.
- “Lightning and Thunder Rounds”



Clinical Findings & Nutrient Connections: Protein



Physical Exam:

- Brittle/fragile hair
- Ridged nails
- Skin changes: darkening, vitiligo
- Underweight and lower muscle mass on BIA
- Delayed wound healing
- Low hand grip strength
- Decreased sit-to-stand & get-up-and-go

Protein Energy Undernutrition (PEU)

Body Region	Clinical Findings (Physical Exam Findings)
Skin	Atrophic, dark, erythematous pigmented macules, edema, exfoliative erythroderma, hyperpigmentation and erythema, loose skin, residual hypopigmentation, shiny, vanished-looking skin, wrinkled, xerosis (dry skin), xerotic crazy paving skin
Hair	Alopecia, brittle/fragile hair, depigmentation of hair and skin (vitiligo), flag sign (bands of depigmentation), hairs plucked easily, hypertrichosis (excessive hair growth), lightening of hair (depigmentation), reddish-brown (rusty) coloring, telogen effluvium, thinning
Nails	Beau's lines, brittle nails, hapalonychia (eggshell nails – soft, thin, bend or break easily), koilonychia (spoon-shaped nails)
Mouth	Caries (cavities), decreased resistance to oral pathogens, degeneration of the dentition supporting structures, delayed exfoliation of primary teeth, delayed wound healing, enamel hypoplasia, loss of buccal fat, poor structural integrity of the dentition, salivary gland atrophy, salivary gland hypofunction

References

PEU CLINICAL FINDINGS

Skin:

- Dipasquale V, Cucinotta U, Romano C. Acute malnutrition in children: pathophysiology, clinical effects and treatment. *Nutrients*. 2020;12(8):2413. doi:10.3390/nu12082413
- Finner AM. Nutrition and hair: deficiencies and supplements. *Dermatol Clin*. 2013;31(1):167-172. doi:10.1016/j.det.2012.08.015
- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003
- Gade A, Hwang JR, Hoegler K, Khan S, Khachemoune A. Therapeutic use of trace elements in dermatology. *Altern Ther Health Med*. 2023;29(4):246-252.
- O'Connor K, Goldberg LJ. Nutrition and hair. *Clin Dermatol*. 2021;39(5):809-818. doi:10.1016/j.clindermatol.2021.05.008

References

PEU CLINICAL FINDINGS

Hair:

- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Gade A, Hwang JR, Hoegler K, Khan S, Khachemoune A. Therapeutic use of trace elements in dermatology. *Altern Ther Health Med*. 2023;29(4):246-252.
- Castelo-Soccio LA. Hair manifestations of systemic disease. *Curr Probl Pediatr Adolesc Health Care*. 2012;42(8):198-203. doi:10.1016/j.cppeds.2012.02.002
- Finner AM. Nutrition and hair: deficiencies and supplements. *Dermatol Clin*. 2013;31(1):167-172. doi:10.1016/j.det.2012.08.015
- Trüeb RM. Effect of ultraviolet radiation, smoking and nutrition on hair. *Curr Probl Dermatol*. 2015;47:107-120. doi:10.1159/000369411
- O'Connor K, Goldberg LJ. Nutrition and hair. *Clin Dermatol*. 2021;39(5):809-818. doi:10.1016/j.clindermatol.2021.05.008
- Coerdts KM, Goggins CA, Khachemoune A. Vitamins A, B, C, and D: a short review for the dermatologist. *Altern Ther Health Med*. 2021;27(4):41-49.
- Capone K, Sentongo T. The ABCs of nutrient deficiencies and toxicities. *Pediatr Ann*. 2019;48(11):e434-e440. doi:10.3928/19382359-20191015-01
- Saleh D, Yarrarapu SNS, Cook C. *Hypertrichosis*. StatPearls Publishing; 2023. Updated August 16, 2023. Accessed June 8, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK534854>
- Yan AC, Jen MV. Skin signs of pediatric nutritional disorders. *Curr Probl Pediatr Adolesc Health Care*. 2012;42(8):212-217. doi:10.1016/j.cppeds.2012.02.003
- Hughes EC, Syed HA, Saleh D. *Telogen Effluvium*. StatPearls Publishing; 2024. Updated May 1, 2024. Accessed June 8, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK430848/>
- Harrison S, Bergfeld W. Diffuse hair loss: its triggers and management. *Cleve Clin J Med*. 2009;76(6):361-367. doi:10.3949/ccjm.76a.08080

References

PEU CLINICAL FINDINGS

Nails:

- Fawcett RS, Linford S, Stulberg DL. Nail abnormalities: clues to systemic disease. *Am Fam Physician*. 2004;69(6):1417-1424.
- Gade A, Hwang JR, Hoegler K, Khan S, Khachemoune A. Therapeutic use of trace elements in dermatology. *Altern Ther Health Med*. 2023;29(4):246-252.
- Muddasani S, Lin G, Hooper J, Sloan SB. Nutrition and nail disease. *Clin Dermatol*. 2021;39(5):819-828. doi:10.1016/j.clindermatol.2021.05.009
- Williams, ME. Examining the fingernails when evaluating presenting symptoms in elderly patients. Medscape Family Medicine. Published November 23, 2009. Accessed May 29, 2025. https://www.medscape.com/viewarticle/712251_1
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Lee DK, Lipner SR. Optimal diagnosis and management of common nail disorders. *Ann Med*. 2022;54(1):694-712. doi:10.1080/07853890.2022.2044511
- Seshadri D, De D. Nails in nutritional deficiencies. *Indian J Dermatol Venereol Leprol*. 2012;78(3):237-241. doi:10.4103/0378-6323.95437
- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003

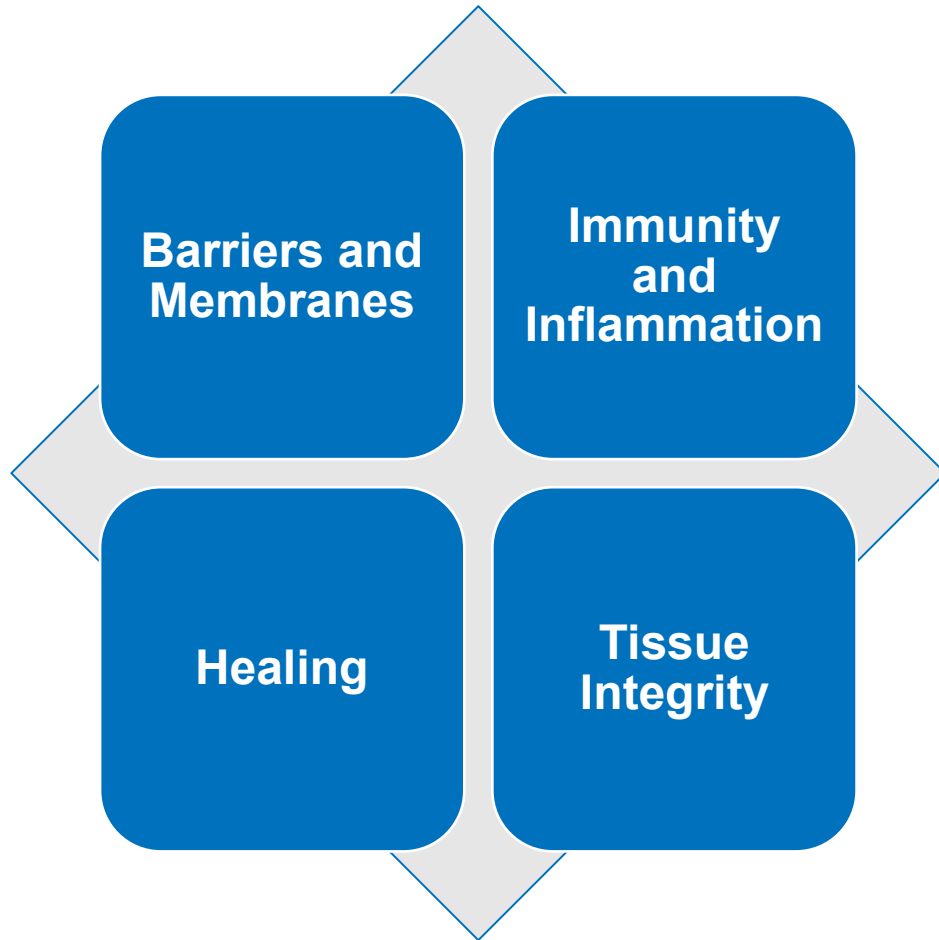
References

PEU CLINICAL FINDINGS

Mouth:

- Mulla SA, Ansari A, Bhattacharjee M, Patil AH, Methal SA. Oral manifestations of nutritional deficiencies: a micro review of macro issues in the mouth. *Oral and Maxillofac Pathol J*. 2023;14(2):217-219.
- Tolkachjov SN, Bruce AJ. Oral manifestations of nutritional disorders. *Clin Dermatol*. 2017;35(5):441-452. doi:10.1016/j.clindermatol.2017.06.009
- Hung M, Patel H, Lee S, Nguyen J, Mohajeri A. The influence of vitamin D levels on dental caries: a retrospective study of the United States population. *Nutrients*. 2024;16(11):1572. doi:10.3390/nu16111572
- Shenkin A, Talwar D, Ben-Hamouda N, et al. LLL 44-1 micronutrients in clinical nutrition: trace elements. *Clin Nutr ESPEN*. 2024;61:369-376. doi:10.1016/j.clnesp.2024.04.007
- Cazzolla AP, Cosola MD, Ballini A, et al. The association between nutritional alterations and oral lesions in a pediatric population: an epidemiological study. *Biomed Res Int*. 2021;2021:9992451. doi:10.1155/2021/9992451
- Dipasquale V, Cucinotta U, Romano C. Acute malnutrition in children: pathophysiology, clinical effects and treatment. *Nutrients*. 2020;12(8):2413. doi:10.3390/nu12082413
- Rahman N, Walls A. Chapter 12: nutrient deficiencies and oral health. *Monogr Oral Sci*. 2020;28:114-124. doi:10.1159/000455379
- Stoopler ET, Villa A, Bindakhil M, Díaz DLO, Sollecito TP. Common oral conditions: a review. *JAMA*. 2024;331(12):1045-1054. doi:10.1001/jama.2024.0953

Clinical Findings & Nutrient Connections: Essential Fatty Acids



Physical Exam:

- Seborrheic Dermatitis
- Acne
- Hyperkeratosis Pilaris
- Phrynoderma
- Xerosis
- Brittle Nails
- Poor Wound Healing
- Immune Dysfunction

EFA Deficiency

Body Region	Clinical Findings (Physical Exam Findings)
Skin	Alopecia, dermatitis (seborrheic-like) changes, desquamation, phrynoderma (aka toad skin; firm, follicular, hyperkeratotic papules, characteristically distributed on the extensor extremities and buttocks; the shoulders, posterior neck, back, and abdomen may also be involved), poor wound healing, scaly, dry rash
Hair	Alopecia, hair loss of scalp and eyebrows, lightening of hair (depigmentation), telogen effluvium

References

EFA DEFICIENCY CLINICAL FINDINGS

Skin:

- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Gade A, Hwang JR, Hoegler K, Khan S, Khachemoune A. Therapeutic use of trace elements in dermatology. *Altern Ther Health Med*. 2023;29(4):246-252.
- Aşkın Ö, Uzunçakmak TKÜ, Altunkalem N, Tüzün Y. Vitamin deficiencies/hypervitaminosis and the skin. *Clin Dermatol*. 2021;39(5):847-857. doi:10.1016/j.clindermatol.2021.05.010
- Galimberti F, Mesinkovska NA. Skin findings associated with nutritional deficiencies. *Cleve Clin J Med*. 2016;83(10):731-739. doi:10.3949/ccjm.83a.15061
- Capone K, Sentongo T. The ABCs of nutrient deficiencies and toxicities. *Pediatr Ann*. 2019;48(11):e434-e440. doi:10.3928/19382359-20191015-01
- Kiani AK, Dhuli K, Donato K, et al. Main nutritional deficiencies. *J Prev Med Hyg*. 2022;63(2 Suppl 3):E93-E101. doi:10.15167/2421-4248/jpmh2022.63.2S3.2752

References

EFA DEFICIENCY CLINICAL FINDINGS

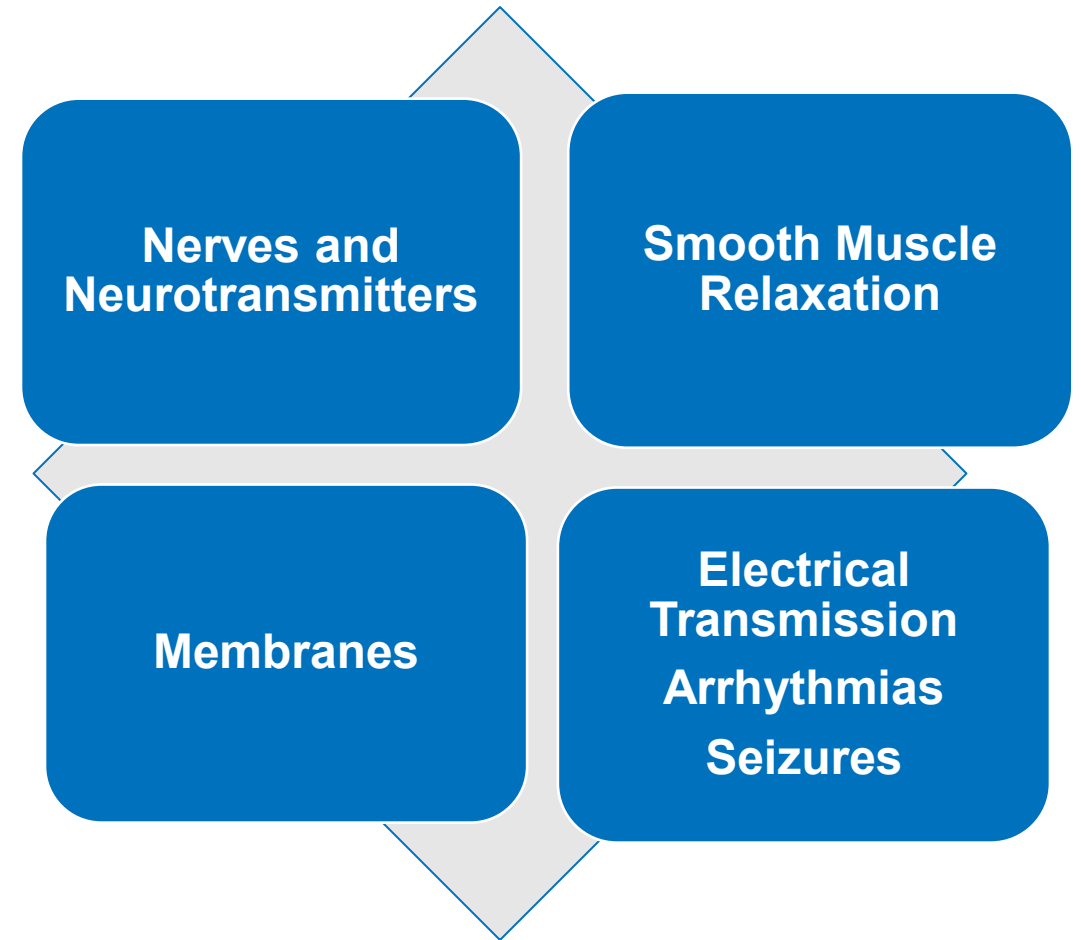
Hair:

- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol*. 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract*. 2019;34(4):490-503. doi:10.1002/ncp.10321
- Gade A, Hwang JR, Hoegler K, Khan S, Khachemoune A. Therapeutic use of trace elements in dermatology. *Altern Ther Health Med*. 2023;29(4):246-252.
- Castelo-Soccio LA. Hair manifestations of systemic disease. *Curr Probl Pediatr Adolesc Health Care*. 2012;42(8):198-203. doi:10.1016/j.cppeds.2012.02.002
- Finner AM. Nutrition and hair: deficiencies and supplements. *Dermatol Clin*. 2013;31(1):167-172. doi:10.1016/j.det.2012.08.015
- Trüeb RM. Effect of ultraviolet radiation, smoking and nutrition on hair. *Curr Probl Dermatol*. 2015;47:107-120. doi:10.1159/000369411
- O'Connor K, Goldberg LJ. Nutrition and hair. *Clin Dermatol*. 2021;39(5):809-818. doi:10.1016/j.clindermatol.2021.05.008
- Yan AC, Jen MV. Skin signs of pediatric nutritional disorders. *Curr Probl Pediatr Adolesc Health Care*. 2012;42(8):212-217. doi:10.1016/j.cppeds.2012.02.003
- Hughes EC, Syed HA, Saleh D. *Telogen Effluvium*. StatPearls Publishing; 2024. Updated May 1, 2024. Accessed June 8, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK430848/>
- Harrison S, Bergfeld W. Diffuse hair loss: its triggers and management. *Cleve Clin J Med*. 2009;76(6):361-367. doi:10.3949/ccjm.76a.08080

Clinical Findings & Nutrient Connections: Magnesium

Physical Exam:

- Muscle spasms and tremors
- Hyperactive reflexes
- Carpopedal spasms
- Nystagmus
- Fibrillations
- Vertigo
- Limited range of motion



Magnesium Insufficiency

Symptoms & Physical Exam Findings

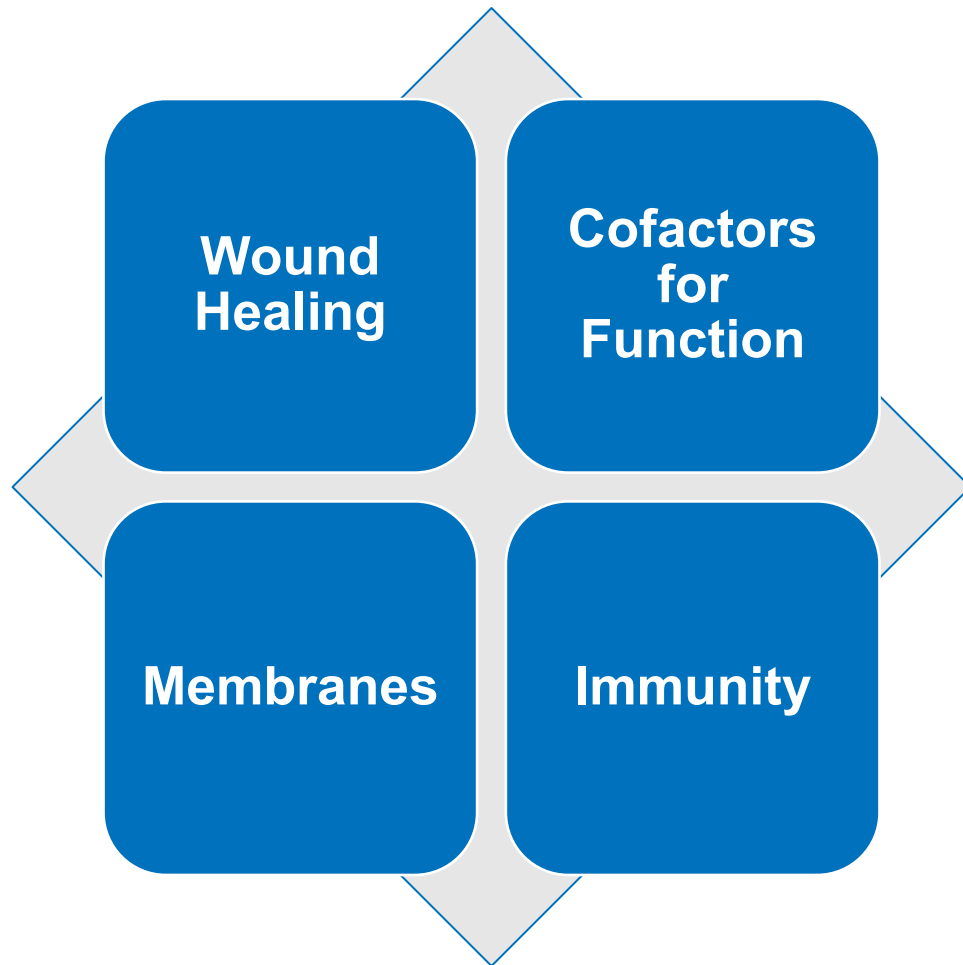
Body Region	Clinical Findings (Physical Exam Findings)
General	Anxiety, lethargy, weakness, agitation, depression, dysmenorrhea, hyperactivity, headache, irritability, dysacusis, low stress tolerance, loss of appetite, nausea, sleep disorders, impaired athletic performance.
Musculature	Muscle spasm, cramps in the soles of the feet, leg cramps, facial muscles, masticatory muscles, and calves, carpopedal spasm, back aches, neck pain, urinary spasms, magnesium deficiency tetany.
Nerves/CNS	Nervousness, increased sensitivity of NMDA receptors to excitatory neurotransmitters, migraine, depression, nystagmus, paresthesia, poor memory, seizures, tremor, vertigo
Gastrointestinal	Constipation
Cardiovascular System	Risk of arrhythmias, supraventricular or ventricular arrhythmias, hypertension, coronary spasm, decreased myocardial pump function, digitalis sensitivity, Torsade de pointes, death from heart disease.
Electrolytes	Hypokalemia, hypocalcemia, retention of sodium
Metabolism	Dyslipoproteinemia (increased blood triglycerides and cholesterol), decreased glucose tolerance, insulin resistance, increased risk of metabolic syndrome, disturbances of bone and vitamin D metabolism, resistance to PTH, low circulating levels of PTH, resistance to vitamin D, low circulating levels of 25(OH)D, recurrence of calcium oxalate calculi.
Miscellaneous	Asthma, chronic fatigue syndrome, osteoporosis, hypertension, altered glucose homeostasis.
Pregnancy	Pregnancy complications (e.g., miscarriage, premature labor, eclampsia)

References

MAGNESIUM INSUFFICIENCY

- Gröber U, Schmidt J, Kisters K. Magnesium in prevention and therapy. *Nutrients*. 2015;7(9):8199-8226. doi:10.3390/nu7095388
- Kiani AK, Dhuli K, Donato K, et al. Main nutritional deficiencies. *J Prev Med Hyg*. 2022;63(2 Suppl 3):E93-E101. doi:10.15167/2421-4248/jpmh2022.63.2S3.2752
- Mulla SA, Ansari A, Bhattacharjee M, Patil AH, Methal SA. Oral manifestations of nutritional deficiencies: a micro review of macro issues in the mouth. *Oral Maxillofac Pathol J*. 2023;14(2):217-219.

Clinical Findings & Nutrient Connections: Zinc



Physical Exam:

- Alopecia
- Night blindness
- Loss or change in taste/smell
- Glossitis
- Periorificial rash
- Seborrhea
- Leukonychia/koilonychia
- Poor wound healing

Zinc Functions Are Wide Ranging

- Anti-inflammatory agent
- Gene regulation
- Tightly regulated with ZIP, ZnT, and metallothioneins
- Broad range of physiologic signs of zinc deficiency
- Organ systems clinically affected by zinc deficiency: epidermal, reproductive, gastrointestinal, immune, central nervous, and skeletal



Clinical Findings & Nutrient Connections: Zinc

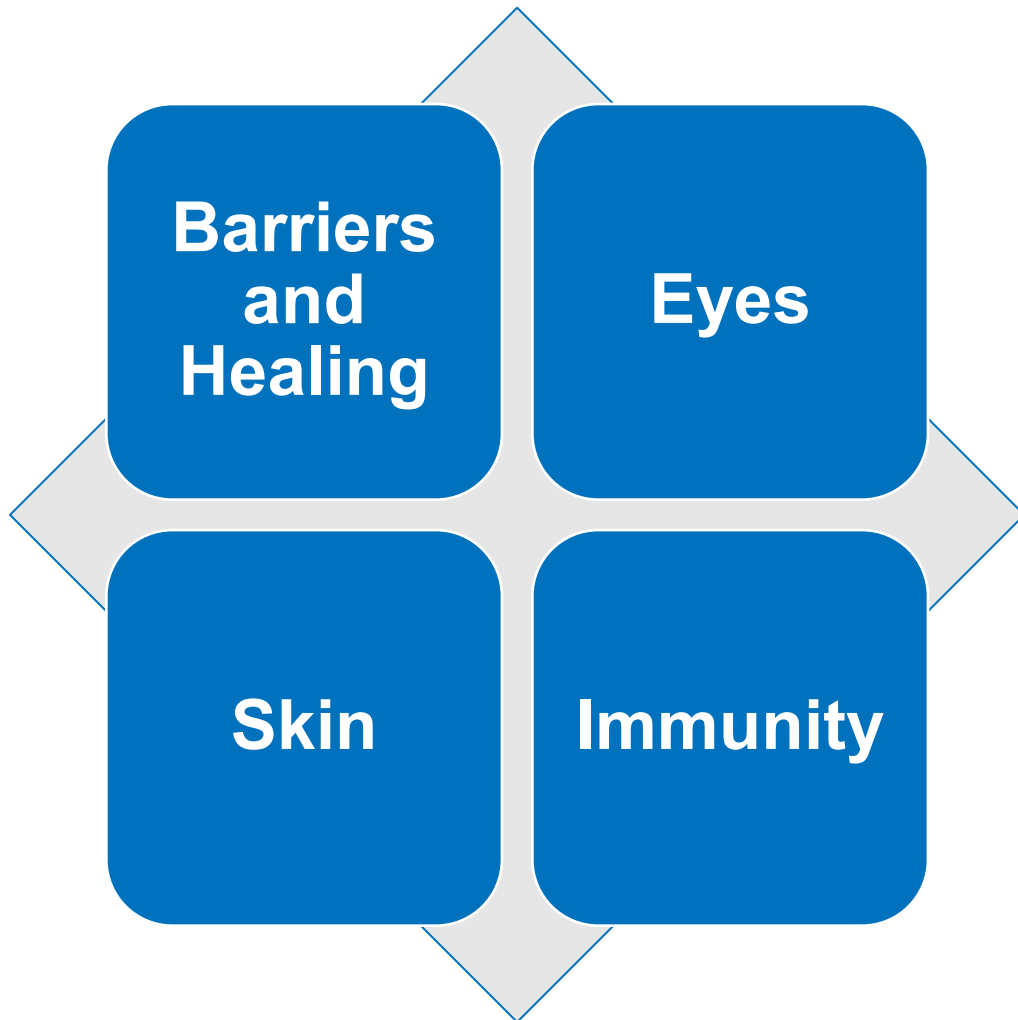
- Hair loss: thinning or alopecia areata
- Acne
- Decreased smell/taste
- Glossitis
- Leukonychia
- Warts/molluscum contagiosum
- Poor wound healing



Park H, Kim CW, Kim SS, Park CW. The therapeutic effect and the changed serum zinc level after zinc supplementation in alopecia areata patients who had a low serum zinc level. *Ann Dermatol.* 2009;21(2):142-146. doi:10.5021/ad.2009.21.2.142. Licensed under CC BY-NC 3.0.



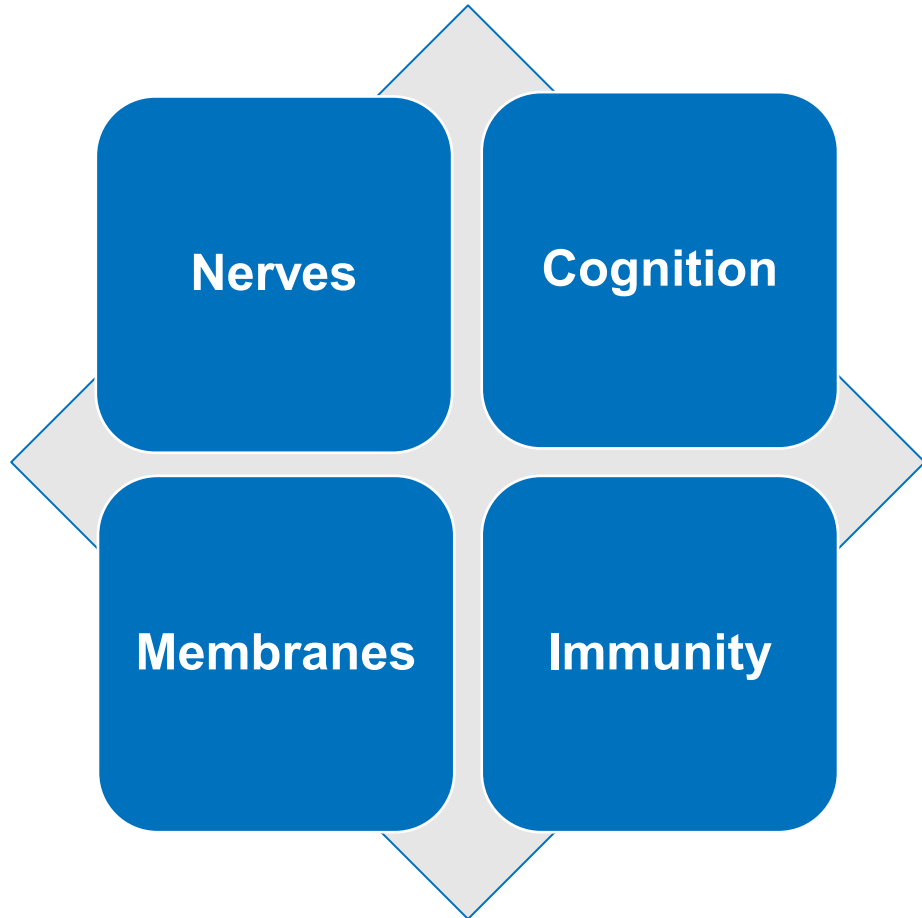
Clinical Findings & Nutrient Connections: Vitamin A



Physical Exam:

- Acne (severe)
- Hyperkeratosis pilaris or white papules
- Poor wound healing
- Poor dark adaptation, twilight vision
- Dry eyes
- Thickened skin
- Follicular hyperkeratosis (thighs/back)

Clinical Findings & Nutrient Connections: Vitamin B12



Physical Exam:

- Glossitis
- Skin changes: darkening, vitiligo
- Neuropathy
- Depression, anxiety, cognitive change, dementia

Nutrition Deficiency or Toxicity and Neuropathy

Insufficiency

Nutrient	Neurological Systems
Vitamin B6	Peripheral neuropathy, seizures, depression
Vitamin B3 (Niacin)	Dementia
Biotin	Seizures, depression, sensorineural hearing loss
Vitamin E	Depression, sensory motor neuropathy, weakness
Copper	Convulsive seizures, hypothermia, and vascular complication
B12 (Cobalamin)	Myelopathy, peripheral neuropathy, neuropsychiatric, optic neuropathy, autonomic dysfunction

Toxicity

Nutrient	Neurological Symptoms
Zinc	Lethargy, hallucinations, neurotoxicity

- Kiani AK, Dhuli K, Donato K, et al. Main nutritional deficiencies. *J Prev Med Hyg.* 2022;63(2 Suppl 3):E93-E101. doi:10.15167/2421-4248/jpmh2022.63.2S3.2752
- Schoofs H, Schmit J, Rink L. Zinc toxicity: understanding the limits. *Molecules.* 2024;29(13):3130. doi:10.3390/molecules29133130
- Wong CY, Chu DH. Cutaneous signs of nutritional disorders. *Int J Womens Dermatol.* 2021;7(5Part A):647-652. doi:10.1016/j.ijwd.2021.09.003
- DiBaise M, Tarleton SM. Hair, nails, and skin: differentiating cutaneous manifestations of micronutrient deficiency. *Nutr Clin Pract.* 2019;34(4):490-503. doi:10.1002/ncp.10321

Clinical Focus

- Summary, conclusion, and key takeaways.



Performing a functional nutrition exam may enable the detection of nutrient insufficiencies that mediate physiological dysfunction and not merely nutrient deficiencies that cause frank disease.

The hair, skin, nails, and mouth represent high-yield body tissues for detecting nutrient insufficiencies.



Anthropometrics
& Vital Signs



Biomarkers &
Functional Labs



**Clinical Findings
(Nutrition Exam)**



Diet and Lifestyle
Assessment

Macronutrient Associations With Hair and Scalp Findings

Protein and Fat

Protein	Physical Exam Findings
Protein deficiency	Alopecia, brittle/fragile hair, depigmentation of hair and skin, flag sign (bands of depigmentation), hair plucked easily, hypertrichosis (excessive hair growth), reddish-brown coloring, telogen effluvium, thinning

Fats & Oils	Physical Exam Findings
EFA deficiency	Alopecia, hair loss of scalp and eyebrows, lightening of hair (depigmentation), telogen effluvium

Micronutrient Associations With Hair and Scalp Findings

Minerals

Minerals	Physical Exam Findings
Copper deficiency	Brittle/fragile hair, depigmentation of hair and skin, Menke's disease (kinky hair disease)
Iron deficiency	Alopecia, telogen effluvium
Manganese deficiency	Slowed growth
Selenium deficiency	Alopecia, depigmentation of hair and skin (lightening of hair)
Selenium excess	Alopecia
Zinc deficiency	Alopecia, brittle/fragile hair, telogen effluvium, white hair

Micronutrient Associations With Hair and Scalp Findings

Vitamins

Vitamins	Physical Exam Findings
Vitamin A deficiency	Follicular hyperkeratosis
Vitamin A excess	Alopecia, telogen effluvium
Vitamin D deficiency	Alopecia
Niacin (B3) deficiency	Alopecia
Biotin (B7) deficiency	Alopecia, loss of hair on scalp, eyebrows, and eyelashes, trichorrhexis nodosa (nodular ball and socket deformities)
Folate (B9) deficiency	Depigmentation (lightening of hair)
Vitamin B12 deficiency	Depigmentation (lightening of hair), gray hair
Vitamin C deficiency	Corkscrew hairs, curling, follicular hyperkeratosis, follicular plugging, perifollicular hemorrhage, swan neck hairs

See: "References: Hair and Scalp Findings and Their Nutrient Associations"

Tongue Physical Exam Signs

Correlation With Specific Nutrient Deficiencies

Nutrient		Physical Exam Findings
Protein		
Protein deficiency		Glossitis (atrophic)

Nutrient		Physical Exam Findings
Minerals		
Iron deficiency		Atrophic glossitis (smooth, glossy, red), burning mouth syndrome, oral candidiasis, geographic tongue
Magnesium deficiency		Burning mouth (symptoms) – glossopyrosis
Zinc deficiency		Burning mouth syndrome, glossitis, geographic tongue

Tongue Physical Exam Signs

Correlation With Specific Nutrient Deficiencies

Nutrient		Physical Exam Findings
Fat-Soluble Vitamins		
Vitamin A deficiency		Oral fungal infections (on tongue and mucosa), thickened tongue, geographic tongue
Vitamin D deficiency		Oral malignancies (mucosal, esophageal, pharyngeal), geographic tongue
Water-Soluble Vitamins		
Riboflavin (B2) deficiency		Atrophic glossitis (smooth, glossy, red), burning mouth syndrome
Niacin (B3) deficiency		Atrophic glossitis (smooth, glossy, red), grey pseudomembrane on tongue, swelling and redness of tongue
Vitamin B6 deficiency		Burning mouth, glossitis, sore tongue, geographic tongue, discoloration of tongue
Folate (B9) deficiency		Atrophic glossitis (smooth, glossy, red), atrophic papilla, burning mouth syndrome, Hunter's glossitis, oral fungal infections, geographic tongue, oral malignancies (mucosal, esophageal, pharyngeal)
Vitamin B12 deficiency		Atrophic glossitis (smooth, glossy, red), burning mouth, fissuring of tongue, geographic tongue, Hunter's glossitis, lingual linear lesions
Vitamin C		Oral fungal infections

See: "References: Tongue Physical Exam Signs: Correlation With Specific Nutrient Deficiencies"

Minerals Associated With Oral Mucosa Findings

Minerals	Physical Exam Findings
Iron deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, oral candidiasis, oral ulcers, pallor of oral mucosa
Zinc deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, periodontitis, xerostomia (dry mouth)

Fat-Soluble Vitamins Associated With Oral Mucosa Findings

Vitamins	Physical Exam Findings
Vitamin A deficiency	Dry lips, oral fungal infections, gingivitis, periodontitis, salivary gland atrophy/hypofunction, xerostomia (dry mouth)
Vitamin A excess	Gingivitis
Vitamin D deficiency	Aphthous stomatitis (canker sores), malignancies (oral, esophageal, pharyngeal), periodontitis, xerostomia (dry mouth)
Vitamin K deficiency	Ecchymosis of buccal mucosa and palate

B Vitamins Associated With Oral Mucosa Findings

Vitamins	Physical Exam Findings
Thiamine (B1) deficiency	Aphthous stomatitis (canker sores), cracked lips
Riboflavin (B2) deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, pharyngeal edema, pharyngitis
Niacin (B3) deficiency	Gingivitis, oral ulcers, xerostomia (dry mouth)
Pyridoxine (B6) deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, gingivitis, oral ulcers, periodontal disease, stomatitis (angular)
Folate deficiency	Aphthous stomatitis (canker sores), burning mouth syndrome, oral candidiasis, gingivitis, stomatitis (angular)
Vitamin B12 deficiency	Aphthous stomatitis (canker sores), hemorrhagic gingivitis, oral ulcers, xerostomia (dry mouth)
Vitamin C deficiency	Black (necrotic) gingiva, oral fungal infections, gum hypertrophy, hemorrhagic gingivitis, periodontal disease, xerostomia (dry mouth)

Vitamins Associated With Skin Findings

Vitamins		Physical Exam Findings
Fat-Soluble		
Vitamin A deficiency		Follicular hyperkeratosis, phrynoderma (toad skin), xerosis
Vitamin A excess		Cheilitis, dermatitis (seborrheic-like), desquamation, pruritic, xerosis
Vitamin D deficiency		Atopic dermatitis, morphea, polymorphous light eruption, psoriasis, skin cancer
Vitamin E deficiency		Phrynoderma (toad skin), xerosis
Vitamin K deficiency		Petechiae, purpura

Vitamins Associated With Skin Findings

Vitamins		Physical Exam Findings	
Water-Soluble			
Riboflavin (B2) deficiency		Angular stomatitis, cheilitis, dermatitis (generalized, atopic, or seborrheic)	
Niacin (B3) deficiency		Casal's necklace, cheilitis, hyperkeratosis, hyperpigmentation, photo-distributed dermatitis	
Vitamin B6		Cheilitis, glossitis, periorificial dermatitis, photo-distributed dermatitis, seborrheic dermatitis	
Biotin (B7)		Dermatitis (eczematous, seborrheic)	
Folate (B9)		Angular cheilitis, hyperpigmentation	
Vitamin B12		Aphthous stomatitis, angular cheilitis, depigmentation, glossitis, hyperpigmentation, seborrheic dermatitis	
Vitamin C		Easy bleeding/bruising, edema (lower extremity), follicular hyperkeratosis, petechiae/purpura, phrynoderma (toad skin), poor wound healing	
Phytonutrients		Physical Exam Findings	
Carotenoid excess		Carotenoderma (yellow-orange discoloration)	

Minerals Associated With Skin Findings

Minerals	Physical Exam Findings
Copper deficiency	Depigmentation
Copper excess	Psoriasis, seborrheic dermatitis
Iron deficiency	Angular stomatitis, cheilitis (angular), pallor, pruritis, psoriasis
Iron excess	Hyperpigmentation
Manganese deficiency	Dermatitis (seborrheic-like)
Selenium deficiency	Acne vulgaris, epidermolysis bullosa (fragile, blistering skin), depigmentation, psoriasis
Zinc deficiency	Acne, acrodermatitis enteropathica, cheilitis, dermatitis (atopic, diaper, seborrheic), poor wound healing, psoriasis, seborrheic dermatitis, xerosis

Nutrient-Associated Nail Findings

Protein	Physical Exam Findings
Protein deficiency	Beau's lines, brittle nails, hapalonychia (eggshell nails), koilonychia (spoon-shaped nails; especially in deficiencies of sulfur-containing amino acids: cysteine or methionine)

Minerals	Physical Exam Findings
Calcium deficiency	Beau's lines, brittle nails, hapalonychia (eggshell nails), longitudinal striations/ridges (onychorrhaxis), onychomadesis, Mee's lines (transverse leukonychia)
Copper excess	Azure (blue) lunula
Iron deficiency	Brittle nails, central ridging, koilonychia (spoon-shaped nails), longitudinal striations/ridges (onychorrhaxis), onycholysis, pallor of nails
Iron excess	Melanonychia (brown or black coloring), splinter hemorrhages of nail bed, Mee's lines (transverse leukonychia)
Magnesium deficiency	Brittle nails
Selenium	Brittle nails, pallor of nails, leukonychia (white nails)
Zinc	Beau's lines, brittle nails, longitudinal striations/ridges (onychorrhaxis), Muehrcke's lines, nail dystrophy, paronychia, Mee's lines (transverse leukonychia)

See: "References: Nail Findings and Their Nutrient Associations"

Nutrient-Associated Nail Findings

Vitamins	Physical Exam Findings
Vitamin A deficiency	Brittle nails, hapalonychia (eggshell nails)
Vitamin A excess	Clubbing
Vitamin D deficiency	Brittle nails, melanonychia (brown or black coloring), hapalonychia (eggshell nails)
Vitamin E	Brittle nails
Riboflavin (B2)	Koilonychia (spoon-shaped nails)
Niacin (B3)	Beau's lines, koilonychia (spoon-shaped nails), longitudinal hyperpigmented streaks, onycholysis, Mee's lines (transverse leukonychia)
Vitamin B6	Hapalonychia (eggshell nails)
Biotin (B7)	Brittle nails
Vitamin B12	Azure (blue) lunula, longitudinal hyperpigmented streaks, melanonychia (brown or black coloring)
Vitamin C	Brittle nails, hapalonychia (eggshell nails), koilonychia (spoon-shaped nails), splinter hemorrhages of nail bed

See: "References: Nail Findings and Their Nutrient Associations"

ABCDs of Nutritional Evaluation

Part C: Clinical Findings-Nutrition Physical Exam

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE